Bernoulli Determinants, Cubic Tornheim Jets, and Canonical Digamma Integrals

Exact harmonic-tail reconstruction, polylogarithmic boundary identities, and finite-part reduction through the seventh power

Research continuation for the ProveIt programme

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Abstract

We develop three exact continuations of the current ProveIt polylogarithm–Hurwitz–Stieltjes programme. First, the conjectured optimal consecutive moment prefix is proved for every unequal even pair of harmonic-tail orders with smaller order 2, 4, or 6. For $(2, 2b)$, $b \geq 2$, the decisive coefficient determinant is $(-1)^{b-1}(2 - B_{2b-2})/(2^b(2b-2)!)$. A rank-two determinant with an arithmetic nonvanishing proof settles the order-four family. A modulo-four Bernoulli congruence and an exact polynomial certificate settle order six. A fixed-size determinant reduction is given for every even pair. For order two, a row of the inverse produces an all-order family of absolutely convergent harmonic and polygamma sums with rational values, together with exact depth-two polylogarithmic boundary identities. Second, every admissible third directional derivative of the Tornheim function at the spectral origin is expressed using two specified higher-depth constants and ordinary zeta derivatives. Two elementary homogeneous conditions characterize formal cancellation of these constants; a three-ray example evaluates exactly. A local holomorphic-kernel theorem extends this structural analysis to every Taylor degree. Third, we give a terminating integration-by-parts normal form for differential polynomials, indexed by partitions whose largest part is repeated. A Gamma-adjoint identity imposes a further parity restriction. With every endpoint constant retained, the canonical sixth digamma moment reduces to the height-one Laurent coefficient λ_5, one mixed finite-part integral, and ordinary zeta data. A seventh-degree identity and a general coincident-polygamma product formula complete the explicit applications. All uniform statements have proofs. Exact finite certificates and independent high-precision diagnostics are supplied separately. The remaining unequal-order moment-prefix conjecture for $8 \leq p < q$, arithmetic elimination of the remaining mixed constants, and the manuscript's S_6 and revised S_8 identities remain open.

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1 Scope, provenance, and the exact questions continued

The starting point is the current collective manuscript *Polylogarithms and their Arithmetic Bridges* and the six archives in the repository's incoming directory at the title-page commit [1]. The source was obtained as a pinned Git checkout; the incoming archives were extracted into separate directories outside that checkout. The research package is an addition for later integration. It does not change the canonical manuscript or the incoming archive contents.

The two newest digamma reports are particularly relevant. *Canonical Digamma and Fifth Moment* proves a finite, but generally longer, initial-moment reconstruction theorem and conjectures that the first q moments already suffice for unequal even orders [2, Sections 7 and 9, `conj:firstq`]. *Higher Digamma Powers and Ordered Gamma Sums* asks for an organized sixth-power reduction and for a differential-polynomial integration calculus [3, Further exact research questions]. The previous coincident/directional-zeta report supplies the complete quadratic ray formula for Tornheim functions, while leaving cubic arithmetic data unresolved [4, Section 6].

1.1 Main contributions and their limits

The first contribution resolves an infinite subfamily of a stated conjecture, rather than adding another finite rank experiment. For the pair $(2, 2b)$, an explicit Padé calculation evaluates the coefficient determinant and a rational extraction row. Classical Bernoulli Hankel machinery is reproved as needed; the mixed matrix perturbation and its application to the moment conjecture are the new continuation [6]. The resulting sums and the polylogarithmic boundary family are identities at every $b \geq 2$. A rank-two version proves the entire $(4, 2b)$ family. A congruence modulo four and a finite polynomial certificate prove $(6, 2b)$ as well. For every even pair $(2a, 2b)$ the remaining determinant is reduced explicitly to size a; the cases $8 \leq p < q$ remain unresolved here.

The second contribution is an exact cancellation theorem for third spectral derivatives. The spectral origin is not a regular point of the three-variable Tornheim function. We first isolate its polar part, then differentiate specified holomorphic ray restrictions. The elementary axis and quasi-shuffle restrictions determine enough of the remaining Taylor coefficients to yield a complete two-coordinate formula and explicit combinations containing ordinary zeta derivatives alone. The formal ambiguity theorem states precisely what these particular local restrictions leave undetermined. It is not a transcendence or arithmetic-independence theorem.

The third contribution answers the algebraic part of the proposed differential-polynomial classification. Its partition basis gives an exact finite algorithm for reduction modulo total derivatives. This is a constructive specialization of classical differential algebra [7], with a new application to the current Gamma-normalized finite-part problem. The normalizations at degrees six and seven are computed explicitly. A remaining mixed integral is displayed, not silently declared independent or replaced by an unevaluated quantity under a new name.

The principal status changes are summarized in Table 1. All references to novelty in this article mean an addition relative to the inspected ProveIt material. A complete historical priority claim would require a broader comparison with all specializations of the classical literature.

1.2 Conventions that affect the constants

For $r \geq 1$, set $H_n^{(r)} = \sum_{k=1}^{n} k^{-r}$ and $H_0^{(r)} = 0$. Bernoulli numbers and polynomials use

$$\frac{te^{xt}}{e^t - 1} = \sum_{n \geq 0} B_n(x) \frac{t^n}{n!}, \qquad B_n = B_n(0), \qquad B_1 = -\frac{1}{2}.$$

Table 1: Source questions and the result established here.

Question	Proved result	Remaining scope
Optimal first-q prefix	Complete proofs for smaller order 2, 4, or 6; a fixed-size determinant for every even pair	Unequal $8 \le p < q$
Higher Tornheim ray derivatives	All cubic directions; exact cancellation criterion and an all-degree local kernel	Arithmetic values of the two cubic coordinates
Differential digamma classification	Canonical partition basis, bidegree dimensions, Gamma parity and endpoints	The image after additional spectral relations
Sixth and seventh moments	Canonical sixth formula with one mixed moment; explicit seventh formula	Spectral evaluation of the mixed integrals
Short Gaussian evaluations	No status change	S_6 and revised S_8

The digamma function is $\psi(x) = \Gamma'(x)/\Gamma(x)$. The generalized Stieltjes convention is

$$\zeta(1+u,a) = \frac{1}{u} + \sum_{j\ge 0} \frac{(-1)^j \gamma_j(a)}{j!} u^j, \qquad \operatorname{Re} a > 0. \tag{1.1}$$

We abbreviate $\gamma = \gamma_0(1)$ and $\gamma_j = \gamma_j(1)$. Derivatives $\zeta^{(j)}(s)$ are always with respect to the spectral variable, not to a Hurwitz shift. Standard Gamma, polygamma, and Hurwitz formulas are used with these normalizations [5].

If f is meromorphic near 0 and regular on $(0,1]$, then

$$\operatorname{FP} \int_0^1 f(x)\,\mathrm{d}x := \operatorname{CT}_{\varepsilon\downarrow 0} \int_\varepsilon^1 f(x)\,\mathrm{d}x. \tag{1.2}$$

The constant term discards negative powers of ε and logarithms but retains the constant in the original coordinate x. In particular, for a meromorphic primitive g,

$$\operatorname{FP} \int_0^1 g'(x)\,\mathrm{d}x = g(1) - \operatorname{CT}_{x=0} g(x). \tag{1.3}$$

This convention is distinct from the centered asymptotic moment continuation in Section 2; the latter is defined explicitly before it is used. Both conventions eventually lead to ordinary convergent identities after their stated subtractions.

2 Centered harmonic moments and the exact Bernoulli transform

We restate the precise inherited moment formulas needed for the new determinant. This also makes the continuation independent of a choice of numerical regularization. Fix an even $q = 2b \ge 4$ and put

$$A_s(n) = \zeta(s) - H_n^{(s)} = \zeta(s, n+1), \qquad x_n = n + \frac{1}{2},$$

$$D_{2,q}(s) = \sum_{n\ge 0} A_2(n) A_q(n) x_n^{-s}, \qquad \operatorname{Re} s > 1 - q. \tag{2.1}$$

The displayed half-plane is the domain of absolute convergence. For later s, the notation means its unique meromorphic continuation. We write $D_r = D_{2,q}(-2r)$ for $r \ge 0$.

2.1 Regularity at every required moment

The centered Hurwitz expansion, obtained from Euler–Maclaurin, is

$$A_s(n) \sim x_n^{1-s} \left\{ \frac{1}{s-1} + \sum_{h\geq 1} \frac{B_{2h}(1/2)(s)_{2h-1}}{(2h)!} x_n^{-2h} \right\}. \tag{2.2}$$

Here s in A_s is a fixed positive even integer, and $(s)_k$ is the rising factorial. The vanishing of all odd $B_{2h+1}(1/2)$ explains the parity of this expansion. Consequently $A_2(n)A_q(n)$ has only powers x_n^{-q-2h}. Subtracting finitely many of them from (2.1) continues the function to any prescribed left half-plane. Its only possible poles occur at $s = 1-q-2h$, which are negative odd integers. Every D_r is therefore regular.

This argument also proves a useful operational rule. If P is an even polynomial and R is the nonnegative-power part of the asymptotic series for $P(x_n)A_2(n)A_q(n)$, then

$$\sum_{n\geq 0} \{P(x_n)A_2(n)A_q(n) - R(x_n)\} = \sum_r [X^{2r}]P(X)\, D_r. \tag{2.3}$$

The left series is absolutely convergent. Each subtracted monomial has continued sum $\zeta(-2h, 1/2) = 0$; this follows either from $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$ or from the odd Bernoulli polynomial at $1/2$. Thus the analytic continuation and the explicitly subtracted ordinary sum have exactly the same constant.

2.2 The fixed formal coordinate system

Write decreasing-index multiple zeta values as

$$\zeta(a, c) = \sum_{n>m\geq 1} \frac{1}{n^a m^c}, \qquad a \geq 2,\ c \geq 1.$$

Define the initial blocks by

$$C_0 = \zeta(2, q-1) + \zeta(q, 1) + \zeta(q+1), \tag{2.4}$$

$$C_j = \zeta(2, q-2j-1) + \frac{1}{2}\zeta(q+1-2j) + \frac{1}{2j}\sum_{h=1}^{j} \binom{2j}{2h} B_{2j-2h}\zeta(q-2h), \quad 1 \leq j < b. \tag{2.5}$$

For $j \geq b$, define $C_j = \sum_{k=0}^{b-1} a_{jk} Z_k$, where

$$Z_0 = \zeta(0) = -\frac{1}{2}, \qquad Z_k = \zeta(2k) \quad (1 \leq k < b), \tag{2.6}$$

and, with $d = 2j+1$, $e = d - q + 2k - 1$,

$$a_{jk} = B_e \left\{ \frac{\binom{d-q+1}{2-2k}}{d-q+1} + \frac{\binom{d-1}{q-2k}}{d-1} \right\}. \tag{2.7}$$

A binomial summand in this formula is omitted unless its lower index is at least 2 and no greater than its upper index. In particular the first summand occurs only when $k = 0$. This is part of the formula, not a convention that may be replaced by $\binom{u}{0} = 1$.

Proposition 2.1 (Inherited two-tail transform)**.** *For every $r, j \geq 0$,*

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) C_j, \tag{2.8}$$

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r. \tag{2.9}$$

Hence all moments lie in the rational span of the q specified formal labels

$$C_0, \dots, C_{b-1}, \qquad Z_0, \dots, Z_{b-1}.$$

These are the $p = 2$ cases of the already proved two-tail formulas reproduced in [2, Section 7]. We give their finite mechanism to fix the cutoffs and constants. Put

$$S_r(N) = \sum_{n=0}^{N-1} (n + 1/2)^{2r} = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) N^{2j+1}.$$

This is an odd polynomial with zero constant term. The exact identity $A_s(k-1) = A_s(k) + k^{-s}$ and finite summation by parts express D_r as the same linear combination of constant terms of

$$E_d(N) = N^d A_2(N) A_q(N) + \sum_{k=1}^{N} k^{d-q} A_2(k) + \sum_{k=1}^{N} k^{d-2} A_q(k) + \sum_{k=1}^{N} k^{d-q-2}. \tag{2.10}$$

The sign of the first term and the inclusive tail cutoff both matter. When $d < q$, the first ordered sum tends to $\zeta(2, q-d)$. For a positive integer exponent m, the exact reduction

$$\sum_{k=1}^{N} k^m A_s(k) = A_s(N) F_m(N) + \sum_{\ell=1}^{N} \frac{F_m(\ell-1)}{\ell^s}, \quad F_m(N) = \frac{B_{m+1}(N+1) - B_{m+1}}{m+1}, \tag{2.11}$$

reduces all remaining sums to finite Bernoulli polynomials and ordinary power sums. Their constant terms are (2.4)–(2.7); at the transition $d = q+1$ the two $-\gamma/2$ contributions cancel the $+\gamma$ from the final harmonic sum in (2.10). The remaining boundary products give the Z_0 column. This is the inherited finite coefficient calculation, not an assumption of arithmetic independence. Finally the standard Bernoulli generating identity inverts the odd-polynomial transform, giving (2.9).

The existing report proves that a longer specified moment prefix has full formal rank. The next section determines the exact first-q minor for every $p = 2$, without any numerical nonvanishing assumption.

3 The optimal moment prefix for every pair $(2, q)$

The source conjecture asks whether the first q centered moments have full formal rank for every pair of even orders $2 \le p < q$. We prove the whole infinite subfamily $p = 2$, with an exact determinant, an explicit rational extraction row, and an ordinary convergent polygamma-tail identity. The order-four and order-six families are proved below; unequal pairs $8 \le p < q$ remain open here.

The Padé and Hankel ingredients used below are classical. In particular, the normalized shifted Bernoulli Hankel determinant is recorded by Dilcher and Jiu [6, §7.3, Eq. (7.12)], who credit the evaluation to Krattenthaler. We reprove the needed formula to make the normalization and the inverse row explicit. The new step relative to the source tail-moment report is the rank-one perturbation calculation giving the factor $2 - B_{q-2}$, and its consequences for the optimal prefix and rational tail sums.

Write $q = 2b$, where $b \ge 2$, and put

$$M_b = (a_{b+t,k})_{0 \le t,k < b}.$$

The entry convention is exactly the source convention: a binomial summand is omitted if its lower index is less than 2. Thus, when $p = 2$, the first binomial summand occurs only in column $k = 0$.

Theorem 3.1 (Exact determinant for the entire order-two family)**.** *For every integer $b \geq 2$,*

$$\det M_b = \frac{(-1)^{b-1}(2 - B_{2b-2})}{2^b(2b-2)!} \neq 0. \tag{3.1}$$

Consequently, for $p = 2$ and $q = 2b$, the first q centered moments $D_0, \ldots, D_{q-1}$ have full rank q on the source's formal coordinate list

$$C_0, \ldots, C_{b-1}, \quad Z_0, \ldots, Z_{b-1}, \qquad Z_k = \zeta(2k), \quad Z_0 = -\tfrac{1}{2}.$$

The determinant of their full coefficient matrix, in this coordinate order, is

$$\frac{(-1)^{b-1}(2 - B_{2b-2})}{2^b(2b-2)!\,(4b-1)!!}. \tag{3.2}$$

This is a formal-rank assertion; it does not assert arithmetic independence of the numerical coordinates.

3.1 A finite Padé calculation, proved explicitly

We use Bernoulli numbers with generating function

$$\frac{t}{e^t - 1} = \sum_{r \geq 0} B_r \frac{t^r}{r!}, \qquad B_1 = -\frac{1}{2},$$

and write

$$\mu_r = \frac{B_{2r}}{(2r)!}, \qquad F(z) = \sum_{r \geq 0} \mu_r z^r = \frac{\sqrt{z}}{2} \coth \frac{\sqrt{z}}{2}.$$

The expression involving $\sqrt{z}$ is an even power series and hence defines a single-valued holomorphic germ at $z = 0$.

Lemma 3.2 (Explicit exponential Padé identity)**.** *For $N \geq 0$, define*

$$P_N(t) = \sum_{k=0}^{N} \frac{(2N-k)!\,N!}{(2N)!\,k!\,(N-k)!}\, t^k.$$

Then

$$e^t P_N(-t) - P_N(t) = \frac{(-1)^N (N!)^2}{(2N)!(2N+1)!}\, t^{2N+1} + O(t^{2N+2}). \tag{3.3}$$

Proof. For $0 \leq r \leq 2N$, the coefficient of t^r in $e^t P_N(-t)$ equals

$$\frac{(2N-r)!}{(2N)!} \sum_{k=0}^{N} (-1)^k \binom{N}{k} \binom{2N-k}{2N-r}.$$

The sum is the coefficient of x^{2N-r} in

$$\sum_{k=0}^{N} (-1)^k \binom{N}{k} (1+x)^{2N-k} = x^N (1+x)^N.$$

It is therefore $\binom{N}{r}$ for $r \leq N$ and zero for $N < r \leq 2N$. These are precisely the coefficients of $P_N(t)$ through degree $2N$. The next coefficient is

$$\frac{1}{(2N)!} \sum_{k=0}^{N} \frac{(-1)^k \binom{N}{k}}{2N+1-k} = \frac{1}{(2N)!} \int_0^1 x^N (x-1)^N \, dx,$$

which gives the stated beta-integral value. □

For $n \geq 0$, take $N = 2n + 1$ and form the polynomials

$$q_n(t^2) = \frac{P_{2n+1}(t) - P_{2n+1}(-t)}{t}, \tag{3.4}$$

$$r_n(t^2) = \frac{P_{2n+1}(t) + P_{2n+1}(-t)}{2}. \tag{3.5}$$

Both have degree n and $q_n(0) = 1$. Direct simplification gives

$$F(z)q_n(z) - r_n(z) = h_n z^{2n+1} + O(z^{2n+2}), \qquad h_n = \frac{((2n+1)!)^2}{(4n+2)!(4n+3)!}. \tag{3.6}$$

Indeed, after setting $z = t^2$, the left side is exactly

$$-\frac{e^t P_{2n+1}(-t) - P_{2n+1}(t)}{e^t - 1}.$$

This proves both the vanishing coefficients and the leading coefficient in (3.6); no normality theorem for Padé approximants is being assumed.

Write

$$q_n(z) = \sum_{\ell=0}^{n} q_{n,\ell} z^\ell, \qquad q_{n,\ell} = \frac{2(4n+1-2\ell)!\,(2n+1)!}{(4n+2)!\,(2\ell+1)!\,(2n-2\ell)!}, \tag{3.7}$$

and reverse its coefficients:

$$\pi_n(x) = x^n q_n(1/x).$$

This is a monic polynomial of degree n. Let $\mathcal{L}$ be the linear functional on polynomials specified by $\mathcal{L}(x^r) = \mu_r$. Comparing coefficients in (3.6) proves

$$\mathcal{L}(x^k \pi_n(x)) = 0 \qquad (1 \leq k \leq n), \tag{3.8}$$

$$\mathcal{L}(\pi_n) = c_n, \qquad c_n = \frac{(2n+1)(2n+2)!}{(4n+2)!}, \tag{3.9}$$

$$\mathcal{L}(x^{n+1}\pi_n) = h_n. \tag{3.10}$$

Here (3.9) is the coefficient of z^n in $r_n(z)$.

3.2 The normalized Bernoulli Hankel determinant

Put

$$H_b = (\mu_{i+j})_{0 \leq i,j < b}, \qquad \Delta_n = \det(\mu_{i+j+1})_{0 \leq i,j < n}, \qquad \Delta_0 = 1.$$

The functional $\mathcal{N}(g) = \mathcal{L}(xg)$ makes the monic polynomials $\pi_0, \pi_1, \ldots$ orthogonal by (3.8). Their squared norms are h_j: the polynomial $\pi_j(x) - x^j$ has degree less than j, so

$$\mathcal{N}(\pi_j^2) = \mathcal{L}(x^{j+1}\pi_j) = h_j.$$

The triangular change from monomials to these monic polynomials has determinant 1. Therefore

$$\Delta_n = \prod_{j=0}^{n-1} \frac{((2j+1)!)^2}{(4j+2)!(4j+3)!}. \tag{3.11}$$

In H_b, replace its last row by the linear combination of rows whose coefficients are those of π_{b-1}. This operation leaves the determinant unchanged because π_{b-1} is monic. The resulting last row is $(c_{b-1}, 0, \ldots, 0)$, whence

$$\det H_b = (-1)^{b-1} c_{b-1} \Delta_{b-1}. \tag{3.12}$$

In particular, the Bernoulli Hankel matrix is nonsingular.

There is also a useful analytic explanation of this nonvanishing. Euler's even-zeta identity gives, for $r \geq 1$,

$$(-1)^{r-1}\mu_r = 2\sum_{m\geq 1}(2\pi m)^{-2r}.$$

After conjugation by $\mathrm{diag}(1,-1,1,\ldots)$, the bottom-right $(b-1)$-square block of H_b is the negative of a positive definite moment matrix. Positivity is strict because a nonzero polynomial cannot vanish at the infinitely many distinct points $(2\pi m)^{-2}$. The top-left entry remains 1, so its Schur complement is positive. This yields the sign $(-1)^{b-1}$ and invertibility independently of the exact product. The exceptional initial coefficient $\mu_0 = 1$ is retained; replacing it by a positive Stieltjes mass would change the problem.

3.3 The mixed matrix is a rank-one perturbation

The source coefficient formula, specialized to $p = 2$, is

$$(M_b)_{tk} = \frac{(2b+2t-1)!}{(2b-2k)!}\,\mu_{t+k} + \mathbf{1}_{k=0}\frac{2t+1}{2}B_{2t}. \tag{3.13}$$

Define diagonal matrices

$$R_b = \mathrm{diag}_{0\leq t<b}(2b+2t-1)!, \qquad E_b = \mathrm{diag}_{0\leq k<b}\,\frac{1}{(2b-2k)!},$$

and a column vector $u = (u_t)_{0\leq t<b}$ by

$$u_t = \mu_t\rho_t, \qquad \rho_t = \frac{(2b)!\,(2t+1)!}{2(2b+2t-1)!}.$$

Then

$$M_b = R_b(H_b + ue_0^{\mathsf{T}})E_b, \tag{3.14}$$

where $e_0 = (1,0,\ldots,0)^{\mathsf{T}}$.

Set $n = b-1$. By (3.8) and (3.9), the first row of H_b^{-1} is the coefficient row of π_n/c_n. Consequently

$$e_0^{\mathsf{T}}H_b^{-1}u = \frac{1}{c_{b-1}}\sum_{t=0}^{b-1} q_{b-1,b-1-t}\,\mu_t\rho_t. \tag{3.15}$$

The factorials in (3.7) cancel to give

$$\frac{1}{c_{b-1}}\sum_{t=0}^{b-1} q_{b-1,b-1-t}\mu_t\rho_t = \frac{1}{2b-1}\sum_{t=0}^{b-1}\binom{2b-1}{2t}(2t+1)B_{2t} \tag{3.16}$$

$$= 1 - B_{2b-2}. \tag{3.17}$$

For clarity, the last equality is an elementary Bernoulli identity. If $N = 2b-1 \geq 3$, then

$$\sum_{k=0}^{N}\binom{N}{k}B_k = B_N(1) = 0, \qquad \sum_{k=0}^{N}k\binom{N}{k}B_k = -NB_{N-1}.$$

Removing the $k = 1$ terms and using $B_k = 0$ for odd $k > 1$ shows

$$\sum_{\substack{0\leq k<N\\ k\text{ even}}}(k+1)\binom{N}{k}B_k = N(1-B_{N-1}),$$

as required. The condition $b \geq 2$ matters: this identity is not the corresponding formula for $N = 1$.

The matrix determinant lemma now yields

$$\det(H_b + ue_0^{\mathsf{T}}) = \det H_b(2 - B_{2b-2}).$$

It remains to simplify the factorial factor. Equations (3.11)–(3.12) give

$$\det R_b \det E_b \det H_b = \frac{(-1)^{b-1}}{2^b(2b-2)!}. \tag{3.18}$$

One quick verification, also a proof for every b, is induction: the left side is $1/2$ at $b = 1$ and the quotient of its values at b and $b - 1$ is

$$-\frac{1}{4(b-1)(2b-3)}.$$

This is exactly the quotient on the right side of (3.18).

For completeness, B_{2m} cannot equal 2 for any $m \geq 1$ by an elementary parity argument; the full von Staudt–Clausen theorem is unnecessary. Work in the ring $\mathbb{Z}_{(2)}$ of rational numbers with odd denominator. The Bernoulli recurrence gives

$$(2m+1)(2B_{2m}) = 2m - 1 - \sum_{j=1}^{m-1} \binom{2m+1}{2j} (2B_{2j}).$$

Inductively $2B_{2j} \in \mathbb{Z}_{(2)}$ and is congruent to 1 modulo 2. The binomial sum in the last display is even modulo 2, since

$$\sum_{j=1}^{m-1} \binom{2m+1}{2j} = 2^{2m} - 2m - 2.$$

Thus $2B_{2m}$ is also congruent to 1 modulo 2. In particular its 2-adic valuation is 0, whereas $2 \cdot 2$ is even. This proves $2 - B_{2m} \neq 0$ and completes (3.1). Finally the forward Bernoulli transform has diagonal entries $1/(2r+1)$, giving (3.2).

3.4 An all-order rational extraction identity

The same calculation identifies one entire row of the inverse, without an unspecified matrix inversion.

Theorem 3.3 (Explicit extraction of the rational coordinate)**.** *For $b \geq 2$,*

$$\sum_{t=0}^{b-1} \binom{2b-1}{2t} C_{b+t} = -\frac{(2b-1)(2 - B_{2b-2})}{4}. \tag{3.19}$$

Equivalently, with

$$w_{b,r} = \sum_{\substack{0 \leq t < b \\ r \leq b+t}} \binom{2b-1}{2t} \binom{2b+2t+1}{2r} 2^{-2(b+t-r)}, \qquad 0 \leq r \leq 2b-1, \tag{3.20}$$

the centered moments satisfy

$$\sum_{r=0}^{2b-1} w_{b,r} D_r = -\frac{(2b-1)(2 - B_{2b-2})}{4}. \tag{3.21}$$

Proof. The rank-one inverse formula and (3.17) show that

$$e_0^{\mathsf{T}}(H_b + ue_0^{\mathsf{T}})^{-1} = \frac{e_0^{\mathsf{T}} H_b^{-1}}{2 - B_{2b-2}}.$$

Insert the coefficient formula for q_{b-1} and the diagonal factors in (3.14). The resulting first row is

$$(M_b^{-1})_{0t} = \frac{2}{(2b-1)(2 - B_{2b-2})} \binom{2b-1}{2t}. \tag{3.22}$$

Multiplication by $(C_b, \dots, C_{2b-1})^{\mathsf{T}}$ recovers $Z_0 = -1/2$, which proves (3.19). Substitution of the source inverse Bernoulli transform

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r$$

gives (3.21). □

As an exact check, for $b = 3$ the block identity is

$$C_3 + 10C_4 + 5C_5 = -\frac{61}{24}.$$

Multiplying its moment form by 1024 reproduces the source's $(2, 6)$ identity with coefficients $61, 8204, 116000, 370048, 303360, 56320$ and right side $-7808/3$. The source example is therefore a specialization of an all-order formula, rather than an isolated elimination.

3.5 An ordinary convergent series at every even order

The finite-part identity has a direct, absolutely convergent interpretation with an explicitly specified polynomial subtraction. Set

$$S_b(y) = \frac{y^{2b+1}}{2}\big((1+y)^{2b-1} + (1-y)^{2b-1}\big), \tag{3.23}$$

$$W_b(x) = S_b(x + 1/2) - S_b(x - 1/2) = \sum_{r=0}^{2b-1} w_{b,r} x^{2r}. \tag{3.24}$$

For an even integer $s \geq 2$, define rational coefficients

$$c_{s,0} = \frac{1}{s-1}, \qquad c_{s,h} = \frac{B_{2h}(1/2)}{(2h)!}(s)_{2h-1} \quad (h \geq 1),$$

where $(s)_r$ is a rising factorial. Put

$$T_{b,h} = \sum_{i=0}^{h} c_{2,i} c_{2b,h-i},$$

and define the explicit even polynomial

$$R_b(x) = \sum_{r=b}^{2b-1} \sum_{h=0}^{r-b} w_{b,r} T_{b,h} x^{2(r-b-h)}. \tag{3.25}$$

Only the finitely many coefficients with $h \leq b - 1$ occur.

Corollary 3.4 (Convergent rational tail sums)**.** *Let $x_n = n + 1/2$ and $A_s(n) = \zeta(s) - H_n^{(s)}$. For every $b \geq 2$,*

$$\sum_{n=0}^{\infty} \{W_b(x_n)A_2(n)A_{2b}(n) - R_b(x_n)\} = -\frac{(2b-1)(2-B_{2b-2})}{4}. \tag{3.26}$$

The summand is $O(n^{-2})$, so the series is absolutely convergent. Equivalently, one may replace the tail product by

$$A_2(n)A_{2b}(n) = \frac{\psi'(n+1)\psi^{(2b-1)}(n+1)}{(2b-1)!}.$$

Proof. The centered Hurwitz expansion gives

$$A_s(n) \sim x_n^{1-s} \sum_{h\geq 0} c_{s,h} x_n^{-2h}, \qquad A_2(n)A_{2b}(n) \sim x_n^{-2b} \sum_{h\geq 0} T_{b,h} x_n^{-2h}.$$

The polynomial R_b in (3.25) is precisely the sum of all nonnegative powers in the product with W_b. After its subtraction the next possible term is $O(x_n^{-2})$. Every removed term has continued sum zero because $\zeta(-2r, 1/2) = 0$ for $r \geq 0$. Thus the absolutely convergent sum equals the finite-part moment combination in (3.21). Alternatively, the partial sum of each removed centered even power is an odd polynomial in the upper integer cutoff with no constant term, which proves the same constant-term equality without interchanging a divergent series. The polygamma expression is the standard positive-even-order identity $A_s(n) = \psi^{(s-1)}(n+1)/(s-1)!$. □

For example, when $b = 2$, multiplication by 64 gives the compact ordinary identity

$$\sum_{n=0}^{\infty} \left\{ (7 + 412x_n^2 + 2000x_n^4 + 1344x_n^6)A_2(n)A_4(n) - 448x_n^2 - \frac{1216}{3} \right\} = -88. \tag{3.27}$$

An independently reconstructed example at $b = 4$ is also useful for checking the size and sign of the subtraction. Define

$$\begin{aligned} \mathsf{P}_4(X) &= 547 + 129756X^2 + 3657584X^4 + 27896512X^6 \\ &\quad + 71997696X^8 + 66311168X^{10} + 20500480X^{12} + 1720320X^{14}, \end{aligned} \tag{3.28}$$

$$\mathsf{R}_4(X) = 245760X^6 + 2334720X^4 + 3955712X^2 + \frac{37765120}{63}. \tag{3.29}$$

These are $2^{14}W_4$ and $2^{14}R_4$, respectively. Then

$$\boxed{\sum_{n=0}^{\infty} \{\mathsf{P}_4(x_n)A_2(n)A_8(n) - \mathsf{R}_4(x_n)\} = -\frac{169984}{3}.} \tag{3.30}$$

The associated exact coefficient matrix has determinant $\det M_4 = -83/483840$ and extraction weights $(1, 21, 35, 7)$. The numerical example is not used to infer the formula: every coefficient above follows from the proved finite polynomials.

What this resolves and what remains open. The optimal consecutive-prefix conjecture is now proved for every even pair with smaller order 2, as well as the diagonal family already settled in the source. The determinant factor $2 - B_{q-2}$ explains why a uniform real-sign argument is misleading: its sign can change as the even Bernoulli numbers grow. The nonzero factor is instead proved arithmetically. The following sections extend the determinant method to $p = 4, 6$ and give an exact smaller determinant for every even pair. Nonvanishing for all unequal pairs $8 \leq p < q$ remains open.

4 A fixed-size determinant for every pair of even orders

The preceding argument has a useful extension even where the final nonvanishing problem remains unsolved. For $p = 2a$, $q = 2b$, with $1 \le a \le b$, the optimal-prefix matrix is an explicitly scaled rank-a perturbation of a shifted Bernoulli Hankel matrix. The Hankel matrix is always nonsingular. Moreover, the remaining determinant has size a, independently of q.

For clarity, the general inherited moment notation is

$$D_{p,q}(s) = \sum_{n\ge 0} A_p(n)A_q(n)(n+1/2)^{-s}, \qquad \operatorname{Re} s > 3 - p - q,$$

continued meromorphically thereafter, and $D_r = D_{p,q}(-2r)$. The same centered-parity argument proves regularity at these even arguments. The forward and inverse Bernoulli transforms (2.8)–(2.9) remain valid, now with C_j the constant term at $N \to \infty$ of

$$N^{2j+1}A_p(N)A_q(N) + \sum_{k=1}^{N} k^{2j+1-q}A_p(k) + \sum_{k=1}^{N} k^{2j+1-p}A_q(k) + \sum_{k=1}^{N} k^{2j+1-p-q}.$$

For $j \ge b$ the inherited formula is $C_j = \sum_{k=0}^{b-1} a_{jk}Z_k$, with $Z_0 = -1/2$ and $Z_k = \zeta(2k)$ for $k \ge 1$. Here the coefficients depend on the indicated pair (p, q). With $d = 2j+1$ and $e = d-p-q+2k+1$, they are zero for $e < 0$, and otherwise

$$a_{jk} = B_e \left\{ \frac{\binom{d-q+1}{p-2k}}{d-q+1} + \frac{\binom{d-p+1}{q-2k}}{d-p+1} \right\}. \tag{4.1}$$

Each summand is omitted unless its lower binomial index lies between 2 and its upper index. These are the exact general formulas of [2, Section 7]; the finite summation argument of Section 2 has the same form. In the formal coordinate order $C_0, \dots, C_{b-1}, Z_0, \dots, Z_{b-1}$, the complete first-$q$ determinant is $\det M_{a,b}/(4b-1)!!$. The new theorems below evaluate or reduce the remaining square minor.

Put

$$\mu_j = \begin{cases} B_{2j}/(2j)!, & j \ge 0, \\ 0, & j < 0, \end{cases} \qquad H_{a,b} = (\mu_{t+k-a+1})_{0\le t,k<b}.$$

The convention $\mu_j = 0$ for $j < 0$ is part of this definition; it is not an analytic continuation of the formula for positive indices.

Lemma 4.1 (Factorization of the general source matrix)**.** *Let $M_{a,b} = (a_{b+t,k})_{0\le t,k<b}$, with the source's convention that a binomial summand is omitted when its lower index is less than 2. Define*

$$R_{tt} = (2b + 2t - 2a + 1)!, \qquad E_{kk} = \frac{1}{(2b-2k)!},$$

$$U_{tk} = \mathbf{1}_{k<a}\, \mu_{t+k-a+1} \frac{(2t+1)!(2b-2k)!}{(2a-2k)!(2b+2t-2a+1)!}.$$

Then

$$M_{a,b} = R(H_{a,b} + U)E, \qquad \operatorname{rank} U = a.$$

Proof. In the source entry formula put $j = b+t$ and $e = 2(t+k-a+1)$. When $e \geq 0$, cancellation of the factorial in B_e gives

$$a_{b+t,k} = \mu_{t+k-a+1} \left\{ \frac{(2b+2t-2a+1)!}{(2b-2k)!} + \mathbf{1}_{k<a} \frac{(2t+1)!}{(2a-2k)!} \right\}.$$

When $e < 0$, both sides vanish by the stated convention. This proves the factorization. Only the first a columns of U are nonzero. Their first a rows form an antitriangular matrix with nonzero antidiagonal, because $\mu_{t+k-a+1} = 0$ for $t+k < a-1$ and $\mu_0 = 1$. Thus its rank is exactly a. □

Theorem 4.2 (Nonsingularity and inertia of all the shifted matrices)**.** *The real symmetric matrix $H_{a,b}$ is nonsingular for every $1 \leq a \leq b$. Its numbers of positive and negative eigenvalues are, respectively,*

$$\begin{cases} (b-h,h), & a = 2h, \\ (h+1, b-h-1), & a = 2h+1. \end{cases}$$

Consequently

$$\operatorname{sgn} \det H_{a,b} = \begin{cases} (-1)^h, & a = 2h, \\ (-1)^{b-h-1}, & a = 2h+1. \end{cases}$$

Proof. Euler's even-zeta evaluation gives

$$\mu_r = (-1)^{r-1}\, 2 \sum_{m \geq 1} x_m^r, \qquad r \geq 1, \qquad x_m = (2\pi m)^{-2}.$$

Conjugate $H_{a,b}$ by $\operatorname{diag}(1,-1,1,\ldots)$ and call the result G. For $t+k \geq a$ its entries are $(-1)^a 2 \sum_{m \geq 1} x_m^{t+k-a+1}$; on $t+k = a-1$ they are $(-1)^{a-1}$, and below that antidiagonal they are zero. Every moment matrix $\left(2 \sum_m x_m^{i+j+s}\right)_{0 \leq i,j < d}$ with $s \geq 1$ is positive definite: its quadratic form is $2 \sum_m x_m^s P(x_m)^2$, and a nonzero polynomial cannot vanish at all the distinct positive points x_m.

First let $a = 2h$. Partition after the first h indices:

$$G = \begin{pmatrix} 0 & B \\ B^{\mathsf{T}} & P \end{pmatrix}.$$

Here P is positive definite. The first h columns of B form an antitriangular square matrix with antidiagonal -1, so B has row rank h. The Schur complement $-BP^{-1}B^{\mathsf{T}}$ is negative definite. Congruence elimination therefore gives $b-h$ positive and h negative eigenvalues.

Now let $a = 2h+1$ and partition after the first $h+1$ indices:

$$G = \begin{pmatrix} ee^{\mathsf{T}} & B \\ B^{\mathsf{T}} & -P \end{pmatrix}, \qquad e = (0,\ldots,0,1)^{\mathsf{T}}.$$

The matrix P is positive definite. In the first h rows of B, its first h columns form an antitriangular matrix with antidiagonal 1. Hence $ee^{\mathsf{T}} + BP^{-1}B^{\mathsf{T}}$ is positive definite: if its quadratic form vanishes, first $e^{\mathsf{T}}v = 0$ and then the indicated minor forces the remaining h components of v to vanish. Congruence elimination gives $h+1$ positive and $b-h-1$ negative eigenvalues. The case $a = b = 1$ is simply $H_{1,1} = (1)$; empty blocks in the argument may otherwise be omitted. Diagonal sign conjugation preserves inertia. □

There is a more explicit reduction than a matrix determinant lemma with an unevaluated $H_{a,b}^{-1}$. Retain the monic polynomials π_n and positive numbers h_n, c_n constructed in the preceding Padé calculation. Write

$$\pi_n(x) = \sum_{t=0}^{n} p_{nt} x^t, \qquad p_{nt} = \frac{2(2n+2t+1)!(2n+1)!}{(4n+2)!(2n-2t+1)!(2t)!}.$$

For $0 \le i, k < a$ set $n_i = b - a + i$, and define the a-square matrix

$$A_{ik} = \sum_{t=0}^{n_i} p_{n_i t} \mu_{t+k-a+1} \left\{ 1 + \frac{(2t+1)!(2b-2k)!}{(2a-2k)!(2b+2t-2a+1)!} \right\}. \tag{4.2}$$

These are finite rational sums with explicitly specified coefficients.

Theorem 4.3 (Reduction to a determinant of size a)**.** *For $1 \le a \le b$,*

$$\det M_{a,b} = (-1)^{a(b-a)} \frac{\prod_{t=0}^{b-1}(2b+2t-2a+1)!}{\prod_{k=0}^{b-1}(2b-2k)!} \left(\prod_{n=0}^{b-a-1} h_n \right) \det A. \tag{4.3}$$

All displayed scalar factors are nonzero. Thus the remaining optimal-prefix nonvanishing question is exactly $\det A \neq 0$*. This reduction by itself does not assert that nonvanishing.*

Proof. Let $\mathcal{L}_a(x^r) = \mu_{r-a+1}$ and recall $\mathcal{N}(x^r) = \mu_{r+1}$. Then $\mathcal{L}_a(x^a g) = \mathcal{N}(g)$. Replace row n of $H_{a,b} + U$ by the linear combination whose coefficients are those of π_n, simultaneously for $0 \le n < b$. This is a triangular row transformation with diagonal 1.

For a column $k \ge a$ the new entry is

$$\mathcal{L}_a(x^k \pi_n) = \mathcal{N}(x^{k-a} \pi_n).$$

It vanishes when $k - a < n$, and equals h_n when $k - a = n$. The last a rows consequently vanish in the last $b - a$ columns, and their first a columns are exactly A. The first $b - a$ rows in the last $b - a$ columns form an upper triangular matrix with diagonal $h_0, \dots, h_{b-a-1}$. Moving the first a columns past the other $b - a$ columns gives the sign $(-1)^{a(b-a)}$. Taking determinants and then restoring R and E proves the formula. □

For reference, the same reduction gives a fully explicit small determinant for the unperturbed Hankel matrix. Put

$$f_\ell(n) = \frac{1}{(2\ell+1)!} \prod_{r=0}^{2\ell-1} (2n - r)(2n + 3 + r), \qquad f_0(n) = 1.$$

Then

$$\det H_{a,b} = (-1)^{a(b-a)} \left(\prod_{n=0}^{b-a-1} h_n \right) \left(\prod_{i=0}^{a-1} c_{n_i} \right) \det\big(f_{a-1-k}(n_i)\big)_{0 \le i,k < a}. \tag{4.4}$$

Indeed,

$$\mathcal{L}_a(x^k \pi_n) = [z^{n+k-a+1}] F(z) q_n(z) = [z^{n+k-a+1}] r_n(z) = c_n f_{a-1-k}(n),$$

where a negative coefficient index means zero. The factorial coefficient formula for r_n proves the last equality, including the zero when $n < a - 1 - k$. In particular, the inertia theorem gives the exact sign statement

$$(-1)^{a(a-1)/2} \det\big(f_{a-1-k}(b - a + i)\big)_{0 \le i,k < a} > 0.$$

A concrete route for the remaining even orders. For each fixed a, the entries in (4.2) can be reduced to finitely many nearby Bernoulli numbers by weighted binomial Bernoulli identities. More explicitly, if $n = b - a + i$, $\ell = a - 1 - k$, and $N = 2(n - \ell) + 1 \ge 1$, the perturbation part of A_{ik} is

$$\frac{2(2n+1)!(2b-2k)!}{(4n+2)!(2a-2k)!N!} \sum_{r=0}^{n-\ell} \binom{N}{2r} B_{2r} (2r + 2\ell + 1)(2n + 2\ell + 2r + 1)^{\underline{2i}}.$$

The underline denotes a falling factorial. The polynomial weight has degree $2i+1 \le 2a-1$. Expressing it in the basis $(N-2r)^{\underline{d}}$ and using

$$\sum_{j=0}^{N} \binom{N}{j} B_j (N-j)^{\underline{d}} = N^{\underline{d}} B_{N-d}(1)$$

therefore reduces it to Bernoulli numbers of orders at most $2a-1$ below N, with the omitted $j=1$ term restored explicitly. When $n < \ell$ the perturbation entry is zero. The small cases $N = 1, 3, \dots, 2i+1$ require retaining $B_1(1) = 1/2$ and $B_0 = 1$; discarding them as generic odd-index zeros would be an error. This is a finite algebraic program for each p, followed by a genuine nonvanishing problem. The sign theorem for $H_{a,b}$ does not determine the sign or nonvanishing of its perturbation.

5 The optimal prefix for every pair $(4, q)$

The same orthogonal polynomials settle a second infinite unequal-order family. The perturbation now has rank two. Its determinant involves three consecutive even Bernoulli numbers, and their elementary 2-adic parity proves that this determinant never vanishes.

Let $p = 4$, $q = 2b$, and $b \ge 3$. In this section write

$$M_b^{[4]} = (a_{b+t,k})_{0 \le t,k < b}.$$

For $b \ge 4$, put

$$d_b = 4b^2 - 6b - 1, \qquad u = B_{2b-6}, \quad v = B_{2b-4}, \quad w = B_{2b-2},$$

and define

$$\begin{aligned} \Xi_b = {} & 4d_b + (b-2)(2b-5)\{u + (2b-1)w(4-u)\} \\ & + (b-1)(2b-3)\{(2b-1)v^2 - 4(b+1)v\}. \end{aligned} \tag{5.1}$$

Theorem 5.1 (Exact determinant for the entire order-four family)**.** *For every integer $b \ge 4$,*

$$\boxed{\det M_b^{[4]} = -\frac{\Xi_b}{6\,2^b(b-1)(2b-4)!} \neq 0.} \tag{5.2}$$

Moreover $4\Xi_b$ belongs to $\mathbb{Z}_{(2)}$ and is odd there; equivalently, $v_2(\Xi_b) = -2$. At the remaining unequal-order case $b = 3$,

$$M_3^{[4]} = \begin{pmatrix} 0 & 3/4 & 1/4 \\ 5/12 & 2/3 & -1/12 \\ 1 & -3/8 & 1/12 \end{pmatrix}, \qquad \det M_3^{[4]} = -\frac{113}{384}. \tag{5.3}$$

Thus the first q centered moments have full formal rank q for every even pair $(p, q) = (4, 2b)$ with $q > 4$. Their complete coefficient determinant is the determinant in (5.2) or (5.3), divided by $(4b-1)!!$.

5.1 A shifted Hankel matrix and its first two inverse rows

Retain $\mu_n = B_{2n}/(2n)!$ for $n \ge 0$, and put $\mu_{-1} = 0$. Define

$$H_b^{[-1]} = (\mu_{t+k-1})_{0 \le t,k < b}.$$

The source formula gives the exact factorization

$$M_b^{[4]} = \mathcal{R}_b(H_b^{[-1]} + U_b)\mathcal{E}_b, \tag{5.4}$$

where

$$(\mathcal{R}_b)_{tt} = (2b + 2t - 3)!, \qquad (\mathcal{E}_b)_{kk} = \frac{1}{(2b - 2k)!},$$

and U_b is zero outside its first two columns, with

$$(U_b)_{tk} = \mu_{t+k-1}\frac{(2t + 1)!(2b - 2k)!}{(4 - 2k)!(2b + 2t - 3)!}, \qquad k = 0, 1. \tag{5.5}$$

This is precisely where the exclusion of a zero lower binomial index matters: no third perturbation column is present.

Let $\mathcal{L}_{-1}(x^j) = \mu_{j-1}$, with $\mathcal{L}_{-1}(1) = 0$. The shifted functional from the preceding section satisfies

$$\mathcal{L}_{-1}(x^2 g) = \mathcal{N}(g).$$

Thus the explicit monic polynomials π_n are orthogonal for the functional on the right. The Padé coefficient identities already proved give

$$\begin{aligned} \mathcal{L}_{-1}(x\pi_n) &= c_n, \\ \mathcal{L}_{-1}(\pi_n) &= c_n\mathfrak{r}_n, \qquad \mathfrak{r}_n = \frac{(2n - 1)(2n)(2n + 3)(2n + 4)}{6}. \end{aligned} \tag{5.6}$$

The latter formula is valid also at $n = 0$, where both sides vanish. For $n \geq 1$ it follows by taking the coefficient of z^{n-1} in $F(z)q_n(z)$: that coefficient equals the coefficient of z^{n-1} in the explicit numerator $r_n(z)$. The quotient of this numerator coefficient and its leading coefficient c_n is exactly $\mathfrak{r}_n$.

Set

$$n = b - 1, \quad m = b - 2, \quad r_+ = \mathfrak{r}_n, \quad r_- = \mathfrak{r}_m, \quad \delta_b = r_+ - r_- = \frac{2(4b - 3)d_b}{3}.$$

Both c_n and c_m are nonzero, and $\delta_b > 0$ for $b \geq 2$. Replace the last two monomial rows in $H_b^{[-1]}$ by the rows for π_m, π_n. This triangular row change has determinant one, and these rows annihilate columns $2, \ldots, b-1$ by $\mathcal{N}$-orthogonality. Expanding by the first two columns therefore gives

$$\det H_b^{[-1]} = -c_m c_n \delta_b \prod_{j=0}^{b-3} h_j \neq 0. \tag{5.7}$$

The two first rows of its inverse are consequently the coefficient rows of the explicit polynomials

$$V_0(x) = \frac{1}{\delta_b}\left(\frac{\pi_n(x)}{c_n} - \frac{\pi_m(x)}{c_m}\right), \tag{5.8}$$

$$V_1(x) = \frac{1}{\delta_b}\left(-r_-\frac{\pi_n(x)}{c_n} + r_+\frac{\pi_m(x)}{c_m}\right). \tag{5.9}$$

Indeed, the two moments in (5.6) give the required Kronecker values in columns zero and one, and orthogonality kills all remaining columns.

Combining (5.7) with the diagonal factors gives

$$\det \mathcal{R}_b \det \mathcal{E}_b \det H_b^{[-1]} = -\frac{d_b}{6\,2^b(b - 1)(2b - 4)!} \qquad (b \geq 2). \tag{5.10}$$

This factorial simplification can be checked inductively: after division by d_b, the value at $b = 2$ is $-1/24$, and the quotient of successive values is $1/[4(b - 1)(2b - 5)]$.

5.2 The four perturbation contractions

For $j = n, m$ and $k = 0, 1$, put

$$s_{j,k} = \frac{1}{c_j} \sum_{t=0}^{j} [x^t]\pi_j(x)\,(U_b)_{tk}, \qquad s_{+,k} = s_{n,k}, \quad s_{-,k} = s_{m,k}.$$

The explicit factorial coefficients of π_j simplify these contractions to the following finite Bernoulli sums:

$$\begin{aligned}
s_{+,0} &= \frac{2b-2}{12} \sum_{\substack{0\le k\le 2b-4\\ k \text{ even}}} \binom{2b-3}{k} (k+3)(k+2b)(k+2b+1)B_k,\\
s_{+,1} &= \frac{1}{2b(2b-1)^2} \sum_{\substack{0\le k\le 2b-2\\ k \text{ even}}} \binom{2b-1}{k} (k+1)(k+2b-2)(k+2b-1)B_k,\\
s_{-,0} &= \frac{(2b)(2b-1)(2b-4)}{12} \sum_{\substack{0\le k\le 2b-6\\ k \text{ even}}} \binom{2b-5}{k} (k+3)B_k,\\
s_{-,1} &= \frac{1}{2b-3} \sum_{\substack{0\le k\le 2b-4\\ k \text{ even}}} \binom{2b-3}{k} (k+1)B_k.
\end{aligned} \tag{5.11}$$

For completeness, the polynomially weighted Bernoulli identities needed to evaluate these sums are as follows. If $N \ge 3$ is odd,

$$\sum_{k \text{ even}} \binom{N}{k} (k+\alpha)B_k = N\left(\frac{\alpha+1}{2} - B_{N-1}\right). \tag{5.12}$$

If $N \ge 5$ is odd and P is a monic cubic, then

$$\begin{aligned}
\sum_{k \text{ even}} \binom{N}{k} P(k)B_k = \frac{N}{2}P(1) + N\{P(N-1) - P(N)\}B_{N-1}\\
- N(N-1)(N-2)B_{N-3}.
\end{aligned} \tag{5.13}$$

Here and in (5.12) the sum runs over $0 \le k \le N$. To prove the identities, first include the $k = 1$ term, then expand $P(N-y)$ in the falling factorials of $y = N - k$. For $0 \le j \le 3$,

$$\sum_{k=0}^{N} (N-k)_{\underline{j}} \binom{N}{k} B_k = N_{\underline{j}} B_{N-j}(1).$$

When $N \ge 5$ is odd, only $j = 1, 3$ survive. The falling-factorial coefficient at $j = 1$ is $P(N-1) - P(N)$, and the coefficient at $j = 3$ is -1. Removing the $k = 1$ term adds $NP(1)/2$. The linear version is obtained in exactly the same way. Thus these are polynomial identities, not conjectural Bernoulli reductions.

Applying them to (5.11), for $b \ge 4$ one obtains

$$\begin{aligned}
s_{-,0} &= r_-(1 - u/2), & s_{-,1} &= 1 - v,\\
s_{+,0} &= r_+ - \frac{(2b-3)(2b-2)(2b-1)(4b-3)}{3}v\\
&\qquad - \frac{(2b-3)(2b-2)(2b-4)(2b-5)}{12}u,\\
s_{+,1} &= 1 - \frac{2(4b-3)}{b}w - \frac{(b-1)(2b-3)}{b(2b-1)}v.
\end{aligned} \tag{5.14}$$

The separate treatment of $b = 3$ avoids applying (5.12) at $N = 1$, where that formula is not valid.

The rank-two determinant lemma and (5.8)–(5.9) now give

$$\frac{\det(H_b^{[-1]} + U_b)}{\det H_b^{[-1]}} = \frac{(r_+ + s_{+,0})(1 + s_{-,1}) - (r_- + s_{-,0})(1 + s_{+,1})}{\delta_b} = \frac{\Xi_b}{d_b}. \tag{5.15}$$

The last equality is direct polynomial simplification of the four displayed contractions. Together with (5.10), it proves the determinant formula (5.2).

5.3 A uniform arithmetic proof of nonvanishing

The Bernoulli parity argument in the order-two section proves that

$$\widehat{u} = 2u, \qquad \widehat{v} = 2v, \qquad \widehat{w} = 2w$$

are odd elements of $\mathbb{Z}_{(2)}$, since $b \geq 4$ makes all three Bernoulli indices positive and even. Multiplication of (5.1) by four yields

$$\begin{aligned} 4\Xi_b = 16d_b &+ (b-2)(2b-5)\{2\widehat{u} + (2b-1)\widehat{w}(8-\widehat{u})\} \\ &+ (b-1)(2b-3)\{(2b-1)\widehat{v}^2 - 8(b+1)\widehat{v}\}. \end{aligned} \tag{5.16}$$

Modulo two, the first bracket contributes b and the second contributes $b - 1$. Hence

$$4\Xi_b \equiv b + (b-1) \equiv 1 \pmod 2.$$

Therefore $v_2(\Xi_b) = -2$ and $\Xi_b \neq 0$ for every $b \geq 4$. This proves nonvanishing uniformly, without a real-sign assertion or any assumption about Bernoulli numerators beyond the elementary parity already established.

Updated scope of the initial-prefix problem. The optimal first-q prefix is now proved for all even pairs with $p = 2$, all even pairs with $p = 4$, and the diagonal $p = q$ inherited from the source. The next section settles $p = 6$; the final remaining unequal-order range in this article is $8 \leq p < q$. The rank-two calculation suggests investigating the 2-adic structure of the analogous higher-rank Bernoulli contractions, but no general nonvanishing theorem for those contractions is assumed here.

5.4 A convergent harmonic–polygamma identity at orders four and six

The exceptional matrix (5.3) also gives a concrete ordinary convergent identity. Let $x_n = n + \frac{1}{2}$ and $A_s(n) = \zeta(s) - H_n^{(s)}$, and put

$$\begin{aligned} W_{4,6}(x) = 67 &+ 15516x^2 + 324320x^4 + 1434496x^6 \\ &+ 1601280x^8 + 371712x^{10}. \end{aligned}$$

Then

$$\boxed{\sum_{n=0}^{\infty} \left\{ W_{4,6}(x_n) A_4(n) A_6(n) - \frac{123904x_n^2 + 316928}{5} \right\} = -21696.} \tag{5.17}$$

The summand is $O(n^{-2})$, so the convergence is absolute. One may equivalently write the tail product as

$$A_4(n)A_6(n) = \frac{\psi^{(3)}(n+1)\psi^{(5)}(n+1)}{720}.$$

Proof. Direct multiplication of the exact matrix in (5.3) gives

$$(-7, 45, 66)M_3^{[4]} = (339/4, 0, 0), \qquad -7C_3 + 45C_4 + 66C_5 = -\frac{339}{8}.$$

Set $S(y) = 512(-7y^7 + 45y^9 + 66y^{11})$. The inverse Bernoulli transform identifies the polynomial of the resulting centered moment combination as

$$S(x + \tfrac{1}{2}) - S(x - \tfrac{1}{2}) = W_{4,6}(x).$$

The centered Hurwitz expansions give

$$A_4(n)A_6(n) = \frac{1}{15}x_n^{-8} - \frac{7}{60}x_n^{-10} + O(x_n^{-12}).$$

Hence the polynomial in nonnegative powers of x_n in its product with $W_{4,6}$ is exactly $(123904x_n^2 + 316928)/5$. Subtracting it leaves an absolutely summable remainder. The removed polynomial has continued sum zero; equivalently, its partial sum has zero constant term in the integer cutoff, by the centered even-power argument used in Corollary 3.4. The ordinary sum therefore equals the finite-part moment combination, namely $512(-339/8) = -21696$. □

6 The optimal prefix for every pair $(6, q)$

A congruence modulo four, rather than only the parity of doubled Bernoulli numbers, settles the next unequal-order family. All matrices in this section are in the same source normalization as Lemma 4.1.

Lemma 6.1 (A doubled-Bernoulli congruence)**.** *For every integer $m \geq 2$,*

$$2B_{2m} \equiv 1 \pmod 4 \quad \text{in } \mathbb{Z}_{(2)}.$$

Proof. The previously proved Bernoulli parity gives $v_2(B_{2j}) = -1$ for $j \geq 1$. Put $r = 2m \geq 4$. The exact Faulhaber identity is

$$\sum_{k=0}^{3} k^r = 4B_r + \sum_{h=2}^{r+1} \binom{r}{h-1} \frac{4^h}{h} B_{r+1-h}.$$

The $h = 2$ term vanishes because $B_{r-1} = 0$. Every $h \geq 3$ term is zero modulo eight: all Bernoulli numbers involved have valuation at least -1, and $2h - 1 - v_2(h) \geq 3$. The left side is $1 + 2^r + 3^r \equiv 2 \pmod 8$. Thus $4B_r \equiv 2 \pmod 8$, proving the assertion. □

Theorem 6.2 (Order-six optimal prefix)**.** *Let $a = 3$ and $b \geq 4$. Then*

$$v_2\left(\frac{\det(H_{3,b} + U)}{\det H_{3,b}}\right) = -3.$$

In particular $\det M_{3,b} \neq 0$. *Consequently the first q centered moments have full formal rank q for every pair $(p, q) = (6, 2b)$ with $q > 6$.*

Here are explicit finite identities certifying the uniform assertion. They also specify the polynomial used in the proof without an unrecorded fitted formula. For $0 \leq i, \ell \leq 2$, put

$$n = b - 3 + i, \qquad N = 2(n - \ell) + 1, \qquad R = 2b - 4 + 2\ell,$$

and retain $f_\ell(n)$ from (4.4). Define

$$\begin{aligned} F_{i\ell} &= f_\ell(n), \\ C_{i\ell} &= \frac{2R!}{(2n+1)(2n+2)(2\ell+2)!N!}, \\ W_{i\ell}(z) &= (z+2\ell+1)(2n+2\ell+z+1)^{\underline{2i}}, \end{aligned} \tag{6.1}$$

$$\alpha_{i\ell h} = \frac{1}{h!}\sum_{j=0}^{h}(-1)^{h-j}\binom{h}{j}W_{i\ell}(N-j). \tag{6.2}$$

All factorial ratios in $C_{i\ell}$ are interpreted as finite rational products in b: $R-N = 1+4\ell-2i$ is a fixed integer. For independent variables $U_1,\dots,U_5$, define the rational 3×3 matrix

$$\mathscr{A}_{i\ell}(b;U) = 2F_{i\ell} + \frac{C_{i\ell}}{2}\sum_{\substack{1\le h\le 2i+1\\ h \text{ odd}}}\alpha_{i\ell h}N^{\underline{h}}U_{3-i+\ell+(h-1)/2}. \tag{6.3}$$

Thus at most three terms occur in any entry. Put

$$D(b) = 16b^4 - 80b^3 + 80b^2 + 50b + 9, \tag{6.4}$$

$$Q(b) = 512b^6 - 6656b^5 + 27296b^4 - 25200b^3 - 91092b^2 + 217836b - 123741. \tag{6.5}$$

Finally define

$$\mathscr{P}(b;U) = 48(2b-3)D(b)\,\frac{\det\mathscr{A}(b;U)}{\det F(b)}. \tag{6.6}$$

The following are identities in rational functions of the indeterminates, not assertions inferred from evaluations at integer values:

$$\det F(b) = \frac{16}{45}(b-1)(2b-3)(4b-7)(4b-5)(4b-3)D(b), \tag{6.7}$$

$$\mathscr{P}(b;U) \in \mathbb{Z}[b,U_1,\dots,U_5], \tag{6.8}$$

$$\mathscr{P}(b;1,1,1,1,1) = -2(2b-5)Q(b). \tag{6.9}$$

These finite algebraic identities are obtained by expanding the six terms of each 3×3 determinant in (6.6) and cancelling the displayed factors. The accompanying certificate performs the expansion with an indeterminate b, checks membership in the integer polynomial ring, and verifies (6.7) and (6.9) exactly. It records the full polynomial $\mathscr{P}$, which has 22 monomials in $U_1,\dots,U_5$ with integer-polynomial coefficients in b. In particular (6.3)–(6.6) are a finite explicit definition from which every coefficient can be recovered.

Proof of Theorem 6.2. Use the last three orthogonal-polynomial rows in Theorem 4.3, reversing their first three columns so that $\ell = 0,1,2$. Divide row i by c_n. The unperturbed rows are $F_{i\ell}$. For $b \ge 6$, the perturbation contraction is

$$C_{i\ell}\sum_{\substack{0\le j\le N\\ j \text{ even}}}\binom{N}{j}W_{i\ell}(j)B_j.$$

Expand $W_{i\ell}(N-y)$ in falling factorials; its coefficients are (6.2). The weighted Bernoulli identity

$$\sum_{j=0}^{N}\binom{N}{j}(N-j)^{\underline{h}}B_j = N^{\underline{h}}B_{N-h}(1)$$

reduces the full sum. Here $N \geq 2i+3$, so only odd h occur, and every surviving Bernoulli index is positive and even. Removing the $j=1$ term contributes $NW_{i\ell}(1)/2$. Direct factorial cancellation gives

$$C_{i\ell}\frac{N}{2}W_{i\ell}(1) = f_\ell(n).$$

Thus the normalized sum of the unperturbed row and its perturbation is precisely (6.3), at $U_j = 2B_{2b-2j}$. The row and column factors cancel in the quotient, so (6.6) proves

$$8\frac{\det(H_{3,b}+U)}{\det H_{3,b}} = \frac{\mathscr{P}(b; 2B_{2b-2}, 2B_{2b-4}, \ldots, 2B_{2b-10})}{6(2b-3)D(b)}. \tag{6.10}$$

For $b \geq 7$, all five doubled Bernoulli numbers are 1 modulo four by Lemma 6.1. The integer-coefficient assertion (6.8) and (6.9) therefore show that the numerator of (6.10) is 2 modulo four: $2b-5$ is odd, and $Q(b) \equiv 1 \pmod 2$. Its valuation is exactly one. The denominator also has valuation one, since $D(b) \equiv 1 \pmod 2$. Hence the quotient on the left of (6.10) is a 2-adic unit, which gives the claimed valuation -3.

The three remaining cases are exact source-matrix calculations:

b	$\det(H_{3,b}+U)/\det H_{3,b}$	$\det M_{3,b}$
4	7527/1240	2509/5120
5	713039/70280	−713039/6451200
6	2968099/1116360	2968099/2786918400.

Each ratio has valuation -3 and each determinant is nonzero. The general factorization and the previously proved nonsingularity of $H_{3,b}$ now establish the full claim. □

A limit of the valuation pattern. The valuations $-1, -2, -3$ proved for $a = 1, 2, 3$ (with $b \geq 4$ in the $a = 2$ case) do not justify a uniform assertion $v_2(\det(H_{a,b}+U)/\det H_{a,b}) = -a$. Already at $a = 4, b = 9$, an exact calculation gives

$$\frac{\det(H_{4,9}+U)}{\det H_{4,9}} = -\frac{334092867653747}{85606130610}, \qquad v_2 = -1.$$

This contradicts that proposed valuation pattern even in the range $b \geq 2a$. It does not refute the optimal-prefix conjecture; the determinant in this example is nonzero. The unequal-order conjecture remaining after the present results is $8 \leq p < q$.

7 A polylogarithmic boundary identity at every even order

The new binomial annihilator has an exact polylogarithmic realization. We use the one-argument, decreasing-index convention

$$\mathrm{Li}_{p,q}(z) = \sum_{n>m\geq 1} \frac{z^n}{n^p m^q}, \qquad |z| < 1.$$

Thus only the largest summation index carries the variable. This convention is essential: a product of two ordinary polylogarithms would give a different coloring.

Proposition 7.1 (The two-tail generating function)**.** *For integers $p, q \geq 2$ and $|z| < 1$, define*

$$\mathcal{F}_{p,q}(z) = \sum_{n\geq 0} \big(\zeta(p) - H_n^{(p)}\big)\big(\zeta(q) - H_n^{(q)}\big) z^n.$$

Then

$$\mathcal{F}_{p,q}(z) = \frac{1}{1-z}\Big\{\zeta(p)\zeta(q) - \zeta(q)\,\mathrm{Li}_p(z) - \zeta(p)\,\mathrm{Li}_q(z) + \mathrm{Li}_{p,q}(z) + \mathrm{Li}_{q,p}(z) + \mathrm{Li}_{p+q}(z)\Big\}. \tag{7.1}$$

Proof. Put $T_n = (\zeta(p) - H_n^{(p)})(\zeta(q) - H_n^{(q)})$. The exact difference is

$$T_{n-1} - T_n = n^{-p}(\zeta(q) - H_n^{(q)}) + n^{-q}(\zeta(p) - H_n^{(p)}) + n^{-p-q}.$$

Multiplication of the generating series by $1 - z$ and substitution of this difference give the result, using

$$\sum_{n\geq 1} \frac{H_n^{(q)}}{n^p} z^n = \mathrm{Li}_{p,q}(z) + \mathrm{Li}_{p+q}(z).$$

All series converge absolutely and locally uniformly on the open unit disk, so the rearrangements are justified. □

The generating-function identity is a routine consequence of finite harmonic summation. The additional content is the particular all-order cancellation furnished by the determinant theorem. Write

$$\mathcal{E} = z\frac{d}{dz} + \frac{1}{2}.$$

For the polynomial W_b in (3.24), let R_b be the nonnegative-power part of the centered asymptotic Laurent series

$$W_b(X)\zeta(2, X + \tfrac{1}{2})\zeta(2b, X + \tfrac{1}{2}). \tag{7.2}$$

Only finitely many Bernoulli coefficients enter R_b; it is an explicit even polynomial of degree $2b - 2$.

Theorem 7.2 (An all-order polylogarithmic boundary evaluation)**.** *For every integer $b \geq 2$,*

$$\boxed{\lim_{z\uparrow 1} \left\{ W_b(\mathcal{E})\mathcal{F}_{2,2b}(z) - R_b(\mathcal{E})\frac{1}{1-z} \right\} = -\frac{2b-1}{4}(2 - B_{2b-2}).} \tag{7.3}$$

The expression in braces has an absolutely convergent Taylor series on the closed unit disk, including its value at $z = 1$.

Proof. On each compact subset of the open disk, polynomial differentiation of the locally normally convergent generating series gives

$$W_b(\mathcal{E})\mathcal{F}_{2,2b}(z) - R_b(\mathcal{E})\frac{1}{1-z} = \sum_{n\geq 0} \left\{ W_b(n + \tfrac{1}{2})A_2(n)A_{2b}(n) - R_b(n + \tfrac{1}{2}) \right\} z^n.$$

The centered Hurwitz expansion contains only even inverse powers. After the displayed subtraction its coefficients are $O(n^{-2})$. The series therefore converges absolutely and uniformly for $|z| \leq 1$. Evaluation at $z = 1$ is exactly the ordinary convergent identity in Corollary 3.4. □

Equations (7.1) and (7.3) give a finite prescription involving classical polylogarithms, depth-two polylogarithms, and their ordinary derivatives. The subtraction is determined before the limit is taken. There is consequently no unspecified regularization constant and no need to assign values to the divergent pieces separately.

8 Third derivatives of Tornheim functions: exact cancellation

Write

$$T(A, B, C) = \sum_{m,n\geq 1} m^{-A}n^{-B}(m + n)^{-C}, \qquad W_{a,b,c}(t) = T(at, bt, ct).$$

The sum is initially absolutely convergent when $\Re(A+C) > 1$, $\Re(B+C) > 1$, and $\Re(A+B+C) > 2$. Thereafter T denotes its meromorphic continuation. The normalization contains no factor 2^t on

the diagonal. The origin is a crossing of polar hyperplanes, so a derivative of $W_{a,b,c}$ is a derivative of the specified one-variable restriction, not a partial derivative of T at the origin.

Borwein–Dilcher and Romik treat the diagonal and related first derivatives; Onodera's work treats functional relations and first and second derivative values in a broader Hurwitz setting [10, 11, 12, 13]. The previous ProveIt continuation already supplies the entire quadratic directional formula and a reduction of the cubic digamma finite part to $W'''_{1,1,1}(0)$. We do not repeat that reduction as a new evaluation. The result here describes every third directional derivative with two specified constants, proves exactly when these two coordinates cancel formally, and gives identities in ordinary zeta derivatives alone. This is a structural continuation of the inspected material; no historical priority over all possible specializations of earlier functional relations is asserted.

Put

$$p = \zeta'(0)\zeta''(0), \qquad q = \zeta'''(0), \qquad r = \zeta'''(-1), \qquad \Omega = W'''_{1,1,1}(0),$$

and define

$$\Phi(a,b,c) = c\{a^2 + b^2 - c(a+b)\}, \tag{8.1}$$

$$\mathscr{Q}(a,b,c) = 3(a+b)(ab+c^2)p - \left\{\frac{a^3+b^3}{2} + 3c^2(a+b) + c^3\right\} q + \left\{c^3 - c\left(\frac{a^3}{b+c} + \frac{b^3}{a+c}\right)\right\} r. \tag{8.2}$$

All these functions are homogeneous of degree three on $\mathscr{D} = \{(a,b,c) : (a+c)(b+c) \neq 0\}$.

Theorem 8.1 (Complete third directional formula)**.** *There is a single direction-independent constant κ, defined explicitly in (8.6) below, such that every $(a,b,c) \in \mathscr{D}$ has a holomorphic restriction at $t = 0$ and*

$$\boxed{W'''_{a,b,c}(0) = abc\{\Omega - 12p + 8q\} + \Phi(a,b,c)\kappa + \mathscr{Q}(a,b,c).} \tag{8.3}$$

Thus all third directional derivatives are in the span of two specified constants over the displayed ordinary-zeta algebra. The assertion is a spanning result. It does not assert arithmetic independence of the two constants or their irreducibility to a smaller constant algebra.

8.1 A holomorphic remainder with a fixed normalization

The pole term needed for the theorem is

$$\mathscr{P}(A,B,C) = -\frac{\Gamma(1-A)\zeta(B-1)}{\Gamma(C)(A+C)} - \frac{\Gamma(1-B)\zeta(A-1)}{\Gamma(C)(B+C)}. \tag{8.4}$$

Lemma 8.2 (Canonical local decomposition)**.** *There is a unique holomorphic germ $J(A,B,C)$, symmetric in A, B, such that*

$$T(A,B,C) = \zeta(A)\zeta(B) + \mathscr{P}(A,B,C) + CJ(A,B,C). \tag{8.5}$$

In particular

$$\kappa = 3J_{AA}(0,0,0) + \frac{\gamma^3}{12} + \frac{\gamma\zeta(2)}{4} + \frac{\zeta(3)}{6} - 3\gamma\zeta''(-1) \tag{8.6}$$

is a well-defined constant.

Proof. Use the classical Mellin representation

$$\Gamma(C)T(A,B,C)=\int_0^\infty x^{C-1}\,\mathrm{Li}_A(e^{-x})\,\mathrm{Li}_B(e^{-x})\,dx$$

and Jonquière's locally normally convergent expansion

$$\mathrm{Li}_A(e^{-x})=\Gamma(1-A)x^{A-1}+\sum_{j\ge0}\frac{(-x)^j}{j!}\zeta(A-j),\qquad 0<x<2\pi,$$

with A in a sufficiently small disc about zero. Subtract from the product the six terms

$$\begin{aligned}S(A,B;x)=\Gamma(1-A)\Gamma(1-B)x^{A+B-2}+\Gamma(1-A)\zeta(B)x^{A-1}+\Gamma(1-B)\zeta(A)x^{B-1}\\ -\Gamma(1-A)\zeta(B-1)x^{A}-\Gamma(1-B)\zeta(A-1)x^{B}+\zeta(A)\zeta(B).\end{aligned}$$

The remainder after multiplication by x^{C-1} is normally integrable on $(0,1)$ near the spectral origin; fixed parameter derivatives add only powers of $\log x$. The integral on $(1,\infty)$ is normally convergent by exponential decay. Integrate the six subtracted terms explicitly. The first three have denominators nonzero at the origin; the last three give $\zeta(A)\zeta(B)/C$ and the two polar terms in (8.4). Thus for a holomorphic symmetric germ H,

$$T=\mathscr{P}+\frac{\zeta(A)\zeta(B)+CH(A,B,C)}{\Gamma(1+C)}.$$

Since $\Gamma(1)=1$, the difference $\zeta(A)\zeta(B)\{1/\Gamma(1+C)-1\}$ is divisible by C as a holomorphic germ. Dividing that difference by C and adding $H/\Gamma(1+C)$ defines J. Uniqueness follows first for $C\neq0$ and then by holomorphy. The initially convergent Mellin calculation and uniqueness of meromorphic continuation justify every equality. □

On a ray, the two pole terms become

$$-\frac{c}{a+c}\frac{\Gamma(1-at)}{\Gamma(1+ct)}\zeta(bt-1)-\frac{c}{b+c}\frac{\Gamma(1-bt)}{\Gamma(1+ct)}\zeta(at-1).$$

These are holomorphic at zero for $(a,b,c)\in\mathscr{D}$, including $c=0$, which proves the removability assertion in Theorem 8.1.

8.2 The two determined second derivatives of the remainder

All derivatives of J in this subsection are at $(0,0,0)$. Let

$$U=J_{AA}=J_{BB},\quad V=J_{AB},\quad X=J_{AC}=J_{BC},\quad Y=J_{CC},\quad M=U+2X.$$

Two elementary restrictions determine M and Y exactly. Counting positive pairs with a prescribed sum gives

$$T(0,0,C)=\zeta(C-1)-\zeta(C).$$

Consequently (8.5) yields

$$CJ(0,0,C)=\zeta(C-1)-\zeta(C)-\frac{1}{6\Gamma(1+C)}-\frac14.$$

Taking the coefficient of C^3 gives

$$Y=\frac{r-q}{3}-\frac{\gamma^3-3\gamma\zeta(2)+2\zeta(3)}{18}.\tag{8.7}$$

For the other restriction, rearrangement of an absolutely convergent double sum, followed by continuation, proves the classical quasi-shuffle identity

$$T(A,0,C)+T(C,0,A)=\zeta(A)\zeta(C)-\zeta(A+C). \tag{8.8}$$

The polar terms in this sum have the holomorphic continuation

$$\begin{aligned}\mathscr{P}(A,0,C)+\mathscr{P}(C,0,A)=&\frac{\Gamma(1-A)\Gamma(1-C)}{12}\frac{\sin(\pi A)+\sin(\pi C)}{\pi(A+C)}\\&-\frac{\zeta(A-1)}{\Gamma(1+C)}-\frac{\zeta(C-1)}{\Gamma(1+A)}.\end{aligned} \tag{8.9}$$

The quotient in the first line is entire locally, since $\sin(\pi A)+\sin(\pi C)=2\sin(\pi(A+C)/2)\cos(\pi(A-C)/2)$. The coefficient of A^2C in $CJ(A,0,C)+AJ(C,0,A)$ is $M/2$. The corresponding coefficient in the first line of (8.9) is $\gamma(\gamma^2+\zeta(2))/24$. Coefficient comparison in (8.8) therefore gives

$$M=p-q+\gamma\zeta''(-1)+(\gamma^2-\zeta(2))\zeta'(-1)-\frac{\gamma^3+\gamma\zeta(2)}{12}. \tag{8.10}$$

This calculation uses the symmetric two-variable identity before specialization; no ambiguous origin value of an individual term is introduced.

8.3 Proof of the complete cubic formula

Let $B(a,b,c)$ be the third derivative at zero of $\zeta(at)\zeta(bt)+\mathscr{P}(at,bt,ct)$. Taylor expansion of the holomorphic remainder gives exactly

$$\begin{aligned}W'''_{a,b,c}(0)&=B(a,b,c)+3c(a^2+b^2)U+6abcV+6c^2(a+b)X+3c^3Y\\&=B(a,b,c)+3\Phi(a,b,c)U+6abcV+3c^2(a+b)M+3c^3Y.\end{aligned} \tag{8.11}$$

For transparent coefficient checking put $z_j=\zeta^{(j)}(-1)$. The ordinary Gamma expansion gives

$$\begin{aligned}B(a,b,c)=&-\frac{a^3+b^3}{2}q+3ab(a+b)p-c\left(\frac{b^3}{a+c}+\frac{a^3}{b+c}\right)r\\&-3c\gamma(a^2+b^2)z_2-3c\{\gamma^2(2ab+c(a+b))+\zeta(2)(2ab-c(a+b))\}z_1\\&+\frac{c}{12}\Big[\gamma^3\{(a+c)^2+(b+c)^2\}+3\gamma\zeta(2)(a^2+b^2-2c^2)\\&\qquad+2\zeta(3)(a^2+b^2-c(a+b)+2c^2)\Big].\end{aligned}$$

For example this follows directly by differentiating

$$\log\frac{\Gamma(1-at)}{\Gamma(1+ct)}=\gamma(a+c)t+\frac{\zeta(2)}{2}(a^2-c^2)t^2+\frac{\zeta(3)}{3}(a^3+c^3)t^3+O(t^4).$$

Substitute (8.7) and (8.10) into (8.11). Collecting the coefficients of Φ and abc gives

$$W'''_{a,b,c}(0)=\Phi(a,b,c)\kappa+abc\{6V-6(\gamma^2+\zeta(2))\zeta'(-1)\}+\mathscr{Q}(a,b,c). \tag{8.12}$$

The diagonal has $\Phi(1,1,1)=0$ and $\mathscr{Q}(1,1,1)=12p-8q$. Hence its specialization identifies the expression in braces as $\Omega-12p+8q$, proving Theorem 8.1.

An optional independent spectral description of the second constant is

$$\kappa=-\frac{1}{2}W'''_{1,0,2}(0)+6p-\frac{41}{4}q+\frac{7}{2}r. \tag{8.13}$$

Here $T(t,0,2t)=Z(2t,t)$ in the strict Euler double-zeta convention. Formula (8.13) is a bridge, not a claimed evaluation of that derivative by ordinary zeta constants.

8.4 All formal cancellations and an explicit exact identity

Corollary 8.3 (Finite combinations cancelling both cubic coordinates)**.** *Let* $\{(a_j, b_j, c_j)\}_{j=1}^{N} \subset \mathscr{D}$ *and* $w_1, \dots, w_N \in \mathbb{C}$*. If*

$$\sum_j w_j a_j b_j c_j = 0, \qquad \sum_j w_j \Phi(a_j, b_j, c_j) = 0, \tag{8.14}$$

then

$$\sum_j w_j W'''_{a_j,b_j,c_j}(0) = \sum_j w_j \mathscr{Q}(a_j, b_j, c_j). \tag{8.15}$$

Conditions (8.14) *are necessary and sufficient for formal cancellation of the two coordinates in Theorem 8.1. Necessity here concerns treating those coordinates as independent symbols; it is not an arithmetic independence theorem for their actual values.*

The locus $\Phi = 0$ supplies especially short two-term identities. For $c \neq 0$ it is $a^2 + b^2 = c(a+b)$. Its asymmetric positive integer point $(3, 6, 5)$ gives

$$\boxed{W'''_{3,6,5}(0) - 90 W'''_{1,1,1}(0) = 81p - \frac{403}{2}q - \frac{245}{11}r.} \tag{8.16}$$

The coefficient $90 = 3 \cdot 6 \cdot 5$ cancels the diagonal coordinate, while $\Phi(3, 6, 5) = 0$ cancels the other one.

For a fixed triple (a, b, c), let $\phi_c = \Phi(a, b, c)$, $\phi_a = \Phi(b, c, a)$, $\phi_b = \Phi(c, a, b)$. A convenient family of weights for the three cyclic rays is

$$(w_c, w_a, w_b) = (\phi_a - \phi_b, \phi_b - \phi_c, \phi_c - \phi_a).$$

Both conditions in (8.14) hold identically. At $(a, b, c) = (1, 2, 3)$ these weights, divided by four, are $(1, 4, -5)$. Thus we obtain the concrete ordinary-zeta identity

$$\boxed{\begin{aligned} &W'''_{1,2,3}(0) + 4W'''_{2,3,1}(0) - 5W'''_{3,1,2}(0) \\ &\qquad = 99\zeta'(0)\zeta''(0) + \frac{207}{2}\zeta'''(0) + \frac{162}{5}\zeta'''(-1). \end{aligned}} \tag{8.17}$$

Every ray in this example has positive coordinates, so no zero-order factor or restriction to a singular hyperplane is involved. Equation (8.17) proves a new explicit cancellation relative to the inspected ProveIt material; it does not evaluate Ω.

8.5 The exact ambiguity of these local identities at every order

The appearance of two coordinates at third order has an algebraic explanation that extends to all orders. This is a statement about what the preceding local restrictions determine, not about how many arithmetically independent constants the actual Tornheim function has.

Theorem 8.4 (Holomorphic ambiguity kernel)**.** *Suppose two meromorphic germs* T_1, T_2 *have the same polar part* (8.4)*, in the precise sense that* $T_i - \mathscr{P}$ *is holomorphic for* $i = 1, 2$*. Suppose both germs satisfy symmetry in their first two variables, have the restriction* $T(A, B, 0) = \zeta(A)\zeta(B)$ *away from the polar set, and satisfy* (8.8)*. Their difference has a unique form*

$$\begin{aligned} \Delta T(A, B, C) = {} & ABC\, S(A, B, C) \\ & + AC(A - C)R(A, C) + BC(B - C)R(B, C), \end{aligned} \tag{8.18}$$

where S *is a holomorphic germ symmetric in* A, B *and* R *is a holomorphic germ symmetric in its two variables. Conversely every expression in* (8.18) *preserves these local identities.*

For homogeneous degree $d \geq 3$, the formal ambiguity space has dimension

$$\left\lfloor \frac{d^2-1}{4} \right\rfloor. \tag{8.19}$$

It vanishes in degrees zero, one and two. Fixing the entire diagonal germ imposes $S(t,t,t) = 0$, and reduces the degree-d dimension by one.

Proof. The common polar term cancels. The common restriction at $C = 0$ then implies $\Delta T = CK(A,B,C)$ for a holomorphic germ K symmetric in A, B. Put $u(A,C) = K(A,0,C)$. The difference of (8.8) says

$$Cu(A,C) + Au(C,A) = 0.$$

At $A = 0$ this gives $u(0,C) = 0$, so $u(A,C) = Av(A,C)$. After division by AC the identity becomes $v(A,C) = -v(C,A)$. An antisymmetric holomorphic germ vanishes on $A = C$ and is uniquely divisible by $A - C$; its quotient is symmetric. Thus $u(A,C) = A(A-C)R(A,C)$ for a unique symmetric germ R. The remainder

$$K(A,B,C) - u(A,C) - u(B,C)$$

vanishes at both $A = 0$ and $B = 0$. It is therefore uniquely divisible by AB, with a holomorphic quotient symmetric in A, B. This quotient is S and proves (8.18). Direct substitution proves the converse and also uniqueness.

In homogeneous degree d, S and R have degree $d-3$. Writing $m = d-3$, a symmetric polynomial in two selected variables and a third unrestricted variable has $\lfloor (m+2)^2/4 \rfloor$ coefficients; a symmetric polynomial in two variables of total degree m has $\lfloor m/2 \rfloor + 1$ coefficients. Their sum is $\lfloor (d^2-1)/4 \rfloor$. On the diagonal the R terms vanish and $\Delta T(t,t,t) = t^3 S(t,t,t)$. At each homogeneous degree evaluation of S at $(1,1,1)$ is a nonzero linear functional, so it removes exactly one dimension. □

At $d = 3$ the two free polynomials in (8.18) are constants; the two direction functions are exactly abc and $\Phi(a,b,c)$. This proves that the cancellation criterion is the complete one obtainable from these local identities alone. At degrees $4, 5, 6$ the corresponding formal dimensions are $3, 6, 8$. Additional functional relations may reduce the arithmetic data further; this theorem does not prejudge that question.

Verification and next questions. The accompanying exact symbolic calculation checks the coefficient collection, the three-ray identity, and the formal kernel dimensions independently by linear algebra. A separate numerical script computes the ray restrictions from the convergent Jonquière–Mellin formula and an incomplete-Gamma tail. Its floating-point residuals are diagnostics and are not used in the proofs. At 50 working decimal digits, the independent comparison for (8.17) gives an absolute residual below 6.3×10^{-33}; both sides are approximately $-451.35351956848625606446741457 45675405$. Concrete next questions are to identify additional functional relations constraining R, to compute the full fourth-order formula in a similarly normalized basis, and to determine whether the two actual cubic constants admit arithmetic relations beyond the formal cancellation proved here.

9 Height-one Stieltjes coefficients and the inherited Gamma bridge

The sixth and seventh reductions use a height-one Laurent coefficient. We record its analytic definition and prove the inherited integral bridge in the exact convention needed here. These

coefficient formulas are established in the incoming fifth-moment report and are related to the height-one series of Young [2, 8]; their inclusion does not constitute a new evaluation of generalized Stieltjes constants.

For depth $d \geq 1$, let

$$Z_d(s) = \zeta(s, \underbrace{1, \dots, 1}_{d-1}) = \sum_{n_1 > \cdots > n_d \geq 1} \frac{1}{n_1^s n_2 \cdots n_d}, \qquad \operatorname{Re} s > 1.$$

Define c_j and $\lambda_{d,j}$ by

$$\frac{1}{\Gamma(1+u)} = \sum_{j \geq 0} c_j u^j, \tag{9.1}$$

$$Z_d(1+u) = \sum_{j=0}^{d} c_j u^{j-d} + \sum_{j \geq 1} \lambda_{d,j} u^j, \qquad \lambda_d := \lambda_{d,1}. \tag{9.2}$$

The second expansion will follow from the next proposition. In particular, $\lambda_{1,j} = (-1)^j \gamma_j / j!$ and $\lambda_1 = -\gamma_1$. The multiple coefficients $\lambda_{d,j}$ have a specified depth; they should not be identified with ordinary Hurwitz coefficients $\gamma_j(a)$ without a further identity.

For $v > 0$, set

$$B(v) = -\log(1 - e^{-v}), \qquad V_d(v) = B(v)^d - \mathbf{1}_{(0,1)}(v)(-\log v)^d,$$

and put

$$I_{d,r} = \int_0^\infty \frac{(\log v)^r V_d(v)}{v}\, dv \quad (r \geq 0).$$

At zero, $V_d(v) = O(v(1 + |\log v|^{d-1}))$; at infinity it decays exponentially. All these integrals converge absolutely.

Proposition 9.1 (Fixed Mellin normalization)**.** *For $d \geq 1$, as a meromorphic identity in $\operatorname{Re} u > -1$,*

$$\Gamma(1+u) Z_d(1+u) = u^{-d} + \frac{u}{d!} \int_0^\infty v^{u-1} V_d(v)\, dv. \tag{9.3}$$

Consequently the coefficients in (9.2) satisfy

$$\lambda_{d,j} = c_{d+j} + \frac{1}{d!} \sum_{r=0}^{j-1} \frac{c_{j-1-r}}{r!} I_{d,r}, \qquad j \geq 1. \tag{9.4}$$

If $\Lambda_d(u) = \Gamma(1+u) Z_d(1+u) - u^{-d}$, then $\Lambda_d^{(j)}(0) = j I_{d,j-1}/d!$ for $j \geq 1$.

Proof. The strict-index generating series with $d-1$ unit indices is $(-\log(1-y))^{d-1}/(d-1)!$. Termwise Mellin integration for $\operatorname{Re} u > 0$ therefore gives

$$\Gamma(1+u) Z_d(1+u) = \frac{1}{(d-1)!} \int_0^\infty \frac{v^u B(v)^{d-1}}{e^v - 1}\, dv = \frac{u}{d!} \int_0^\infty v^{u-1} B(v)^d\, dv.$$

For the second equality integrate by parts, using $B'(v) = -(e^v - 1)^{-1}$; both boundary terms vanish in this half-plane. Subtract the exact integral $\int_0^1 v^{u-1}(-\log v)^d\, dv = d!/u^{d+1}$. The estimates above give holomorphy of the remaining integral for $\operatorname{Re} u > -1$, locally uniformly with every spectral derivative. Expansion at zero and multiplication by (9.1) prove all the coefficient identities. □

This calculation makes the Stieltjes content of the later formulas explicit: the spectral derivative data are ordinary convergent logarithmic Mellin integrals, with the subtraction at $v = 1$ fixed in advance.

Define the Gamma family

$$G(x,t) = \frac{\Gamma(1-t)\Gamma(x)}{\Gamma(x-t)} = \sum_{m\geq 0} C_m(x)\frac{t^m}{m!}. \tag{9.5}$$

At positive integer $x = n$, it is the familiar complete harmonic-star generator $\prod_{k=1}^{n-1}(1-t/k)^{-1}$; this classical connection is also discussed by Sesma [9]. At $x = 0$, the residue of $G(x,t)$ is $-t$. Thus every C_m with $m \geq 2$ has a removable singularity there. Expansion of the constant term gives $C_m(0) = -m!\zeta(m)$, while $G(1,t) = 1$ gives $C_m(1) = 0$.

Proposition 9.2 (Inherited ordinary Gamma integral)**.** *For every $m \geq 2$,*

$$\int_0^1 C_m(x)\,dx = m!(c_m - \lambda_{m-1}). \tag{9.6}$$

The integral is ordinary and the complete integrand is continuous on $[0,1]$.

Proof. Use $F(x,t) = G(x,-t)$ and write $h_1(x) = \psi(x) + \gamma$. The function

$$\Psi(t) = \int_0^1 \{F(x,t) - 1 + th_1(x)\}\,dx$$

is holomorphic near zero. Indeed the residues t/x and $-t/x$ cancel, leaving a jointly regular expression for $x \in [0,1]$ and t in a sufficiently small closed disc. Hence

$$\Psi(t) = \sum_{m\geq 2} \frac{(-1)^m t^m}{m!}\int_0^1 C_m(x)\,dx.$$

For $\operatorname{Re} t > 0$, Euler's beta integral for F and the ordinary digamma integral, combined before integration, yield

$$\Psi(t) = t\int_0^\infty \left\{\frac{(1-e^{-v})^t - 1}{v} + \frac{e^{-v}}{1-e^{-v}}\right\} dv - 1.$$

Fubini is justified for the combined integrand: at zero it is $O(v^{\operatorname{Re} t - 1} + 1)$ after cancellation, and at infinity integration in x leaves an exponentially decaying numerator divided by v. To put this in the fixed subtraction convention, integrate from a lower cutoff δ, use $\int_\delta^\infty (e^v - 1)^{-1}\,dv = -\log(1-e^{-\delta})$, and compare with $\int_\delta^1 (v^t - 1)\,dv/v$. Taking $\delta \downarrow 0$ gives

$$\Psi(t) = t\int_0^\infty \frac{(1-e^{-v})^t - 1 - \mathbf{1}_{(0,1)}(v)(v^t - 1)}{v}\,dv. \tag{9.7}$$

Near zero, $(1-e^{-v})^t - v^t = O(v^{\operatorname{Re} t + 1})$ uniformly on compact parameter sets; at infinity the numerator decays exponentially. Thus this equality continues holomorphically to a neighborhood of zero, and coefficients may be extracted under the integral. Its degree-m coefficient is $(-1)^{m-1}I_{m-1,0}/(m-1)!$. Comparing the two expressions for Ψ gives $\int_0^1 C_m = -mI_{m-1,0}$. Proposition 9.1 then proves (9.6). □

Individual terms in a differential expansion of C_m can have divergent improper integrals even though C_m itself is regular. The finite-part normal form in the next section is what permits those terms to be separated with their exact endpoint contributions retained.

10 Canonical integration by parts for polygamma polynomials

The incoming canonical fifth-moment report asks for an all-degree classification of differential monomials after integration by parts, with their finite endpoint constants retained. The next theorem answers the formal integration-by-parts part of that question. It also gives an exact reduction algorithm and a smaller parity sector for the inverse-Gamma polynomials. The vector-space dimensions below do not count arithmetically independent constants, or independent images under any particular spectral map. Additional special-function identities can reduce the resulting integrals further; the sixth-degree calculation below does exactly this.

Differential-polynomial methods and the exactness of the Euler-operator complex are established tools; see Olver–Shakiban [7]. The elementary construction below is included in full to fix the particular grading, basis, and endpoint convention needed for this application.

10.1 The formal vector-space quotient and a triangular basis

Let

$$\mathscr{R} = \mathbb{Q}[X_1, X_2, \ldots], \qquad D = \sum_{j\geq 1} X_{j+1}\frac{\partial}{\partial X_j}.$$

Each polynomial involves finitely many variables, so this derivation is well defined. Give X_j weight j and factor degree one. Write $\mathscr{R}_{N,\ell}$ for the span of monomials of weight N and factor degree ℓ, and $\mathscr{R}_N = \bigoplus_\ell \mathscr{R}_{N,\ell}$. The derivation raises weight by one and preserves factor degree. The substitution $X_j = \psi^{(j-1)}(x)$ intertwines D with d/dx.

For a partition $\lambda = (\lambda_1 \geq \cdots \geq \lambda_\ell > 0)$, put $X_\lambda = \prod_{i=1}^{\ell} X_{\lambda_i}$. Let $\mathscr{B}_N$ consist of the partitions of N whose largest part occurs at least twice. At weights zero and one, use separately the representatives 1 and X_1.

Theorem 10.1 (Canonical differential-polynomial normal form)**.** *For $N \geq 2$, the classes*

$$\{X_\lambda : \lambda \in \mathscr{B}_N\}$$

are a basis of the vector space $\mathscr{R}_N/D\mathscr{R}_{N-1}$. Every $F \in \mathscr{R}_N$ has a unique decomposition

$$F = \mathcal{N}(F) + D\mathcal{T}(F), \qquad \mathcal{N}(F) \in \operatorname{span}_{\mathbb{Q}}\{X_\lambda : \lambda \in \mathscr{B}_N\}, \quad \mathcal{T}(F) \in \mathscr{R}_{N-1}. \tag{10.1}$$

If $p_\ell(N)$ is the number of partitions of N into exactly ℓ parts, then

$$\dim \frac{\mathscr{R}_{N,\ell}}{D\mathscr{R}_{N-1,\ell}} = p_\ell(N) - p_\ell(N-1), \qquad \dim \frac{\mathscr{R}_N}{D\mathscr{R}_{N-1}} = p(N) - p(N-1). \tag{10.2}$$

The decomposition is produced by a terminating rational algorithm. Every retained monomial uses derivatives of order at most $\lfloor N/2 \rfloor - 1$ after substitution into ψ.

Proof. Order partitions lexicographically by their decreasing lists of parts. Let a monomial have a unique largest part $M \geq 2$. Decrease this part to $M-1$, and call the resulting partition μ. If b is the multiplicity of $M-1$ in μ, the product rule gives

$$DX_\mu = bX_\lambda + R_\lambda. \tag{10.3}$$

Every monomial in R_λ has largest part at most $M-1$: it arises by differentiating one of the smaller factors of X_μ. Thus every such partition is lexicographically smaller than λ. Replace X_λ by $b^{-1}DX_\mu - b^{-1}R_\lambda$. Applying this replacement to the largest remaining unique-largest partition strictly decreases a finite ordered set of partitions of N, so it terminates in the asserted remainder. The replacement preserves factor degree.

For independence, apply D to the partition monomials of weight $N-1$. The leading partition is obtained by increasing one largest part by one; its coefficient is the multiplicity of that largest part and is nonzero. This is a bijection between partitions of $N-1$ and partitions of N with a unique largest part. It also preserves the number of parts. Distinct derivative columns therefore have distinct leading partitions. No nonzero linear combination of those columns can be supported entirely on repeated-largest partitions. This proves both independence and injectivity of D on $\mathscr{R}_{N-1}$, and hence uniqueness of the primitive in (10.1). Counting the columns and rows gives (10.2). Finally $2\lambda_1 \le N$ for each retained partition, giving the derivative-order bound. □

The quotient is a vector-space quotient: $D\mathscr{R}$ is not in general an ideal, so multiplication of classes is not being asserted. The ordinary generating series, including weights zero and one, is

$$\sum_{N\ge 0} \dim(\mathscr{R}_N/D\mathscr{R}_{N-1})q^N = q + \prod_{j\ge 2} \frac{1}{1-q^j}. \tag{10.4}$$

The convention is $D\mathscr{R}_{-1} = 0$. The extra q records X_1, since differentiation kills the constant polynomial. With factor degree recorded by z, the corresponding series is

$$(1-q)\prod_{j\ge 1}(1-zq^j)^{-1} + q.$$

For example, the respective bases at weights four, five, and six are

$$\begin{array}{c|l} 4 & X_1^4,\ X_2^2 \\ 5 & X_1^5,\ X_1X_2^2 \\ 6 & X_1^6,\ X_1^2X_2^2,\ X_2^3,\ X_3^2. \end{array}$$

This explains formally why the fifth digamma calculation introduces $\psi(\psi')^2$ after the scalar power. It does not assert that its integral is independent of the spectral constants already present.

10.2 The exact endpoint functional

For a meromorphic function with a finite principal part at zero, use the original-coordinate finite part

$$\mathrm{FP}\int_0^1 f(x)\,dx = \mathrm{CT}_{\varepsilon\downarrow 0}\int_\varepsilon^1 f(x)\,dx.$$

Here the constant term is taken after all negative powers of ε and all nonconstant powers of $\log\varepsilon$ are removed. For every differential polynomial T,

$$\mathrm{FP}\int_0^1 DT(\psi,\psi',\ldots)\,dx = \mathcal{E}(T) := T(\psi(1),\psi'(1),\ldots) - \mathrm{CT}_{x=0}\,T(\psi(x),\psi'(x),\ldots). \tag{10.5}$$

This follows by the fundamental theorem of calculus before the constant term is taken. In particular a total derivative cannot simply be discarded in the finite-part problem.

The functional is an explicitly finite polynomial calculation. At weight w, use only

$$U_w(x) = -x^{-1} - \gamma + \sum_{k=1}^{w-1}(-1)^{k+1}\zeta(k+1)x^k, \tag{10.6}$$

and substitute $X_j = U_w^{(j-1)}(x)$ to obtain the constant term. Terms omitted from ψ start at x^w; in a weight-w monomial, any occurrence of an omitted term has positive total exponent even when all other factors contribute their leading poles. Thus it cannot affect the constant coefficient. The endpoint at one uses

$$X_1 = -\gamma, \qquad X_j = (-1)^j(j-1)!\zeta(j) \quad (j\ge 2).$$

Consequently $\mathcal{E}(T) \in \mathbb{Q}[\gamma, \zeta(2), \ldots, \zeta(w)]$ for every $T \in \mathscr{R}_w$.

For a formula with entirely finite summation ranges, define $\beta_1 = -\gamma$ and $\beta_j = (-1)^j(j-1)!\zeta(j)$ for $j \geq 2$. Set

$$\alpha_{j,-j} = (-1)^j(j-1)!, \quad \alpha_{1,0} = -\gamma, \quad \alpha_{j,k} = (-1)^{j+k}\frac{(j+k-1)!}{k!}\zeta(j+k) \quad (k \geq 0,\ j+k \geq 2).$$

All other negative-index coefficients are zero. For a partition λ of w,

$$\mathcal{E}(X_\lambda) = \prod_i \beta_{\lambda_i} - \sum_{\substack{k_i \in \{-\lambda_i, 0, 1, \ldots, w-\lambda_i\} \\ k_1 + \cdots + k_\ell = 0}} \prod_i \alpha_{\lambda_i, k_i}. \tag{10.7}$$

This supplies every endpoint constant at arbitrary degree without leaving an unspecified finite-part operation.

The normal-form theorem therefore becomes the exact analytic identity

$$\mathrm{FP}\int_0^1 F(\psi, \psi', \ldots)\, dx = \sum_{\lambda \in \mathscr{B}_N} [X_\lambda]\mathcal{N}(F)\ \mathrm{FP}\int_0^1 \prod_i \psi^{(\lambda_i - 1)}(x)\, dx + \mathcal{E}(\mathcal{T}(F)). \tag{10.8}$$

Its proof uses finite algebra and (10.5); it does not interchange divergent integrals of individual terms.

10.3 Inverse-Gamma polynomials and the parity reduction

Define formal differential polynomials by

$$A_0 = 1, \qquad A_{m+1} = X_1 A_m - DA_m. \tag{10.9}$$

On substituting $X_j = \psi^{(j-1)}$, these become

$$A_m(x) = (-1)^m \Gamma(x)\frac{d^m}{dx^m}\frac{1}{\Gamma(x)}, \qquad \frac{\Gamma(x)}{\Gamma(x-t)} = \sum_{m \geq 0} A_m(x)\frac{t^m}{m!}.$$

Write $N_m = \mathcal{N}(A_m)$ and $T_m = \mathcal{T}(A_m)$, with $N_0 = 1, T_0 = 0, N_1 = X_1, T_1 = 0$.

Theorem 10.2 (Factor parity and an explicit all-degree recurrence)**.** *Every monomial of N_m has factor degree congruent to m modulo two. Thus its number of possible canonical coordinates is at most*

$$r_m = \sum_{\ell \equiv m\ (2)} (p_\ell(m) - p_\ell(m-1)) \qquad (m \geq 2). \tag{10.10}$$

For $m \geq 1$, the complete pair (N_{m+1}, T_{m+1}) is computed from (N_m, T_m) by setting

$$(R, S) = (\mathcal{N}, \mathcal{T})(X_1 N_m - X_2 T_m)$$

and then

$$N_{m+1} = R, \qquad T_{m+1} = S + X_1 T_m - N_m - DT_m. \tag{10.11}$$

All coefficient operations are finite and rational.

Proof. Here is a polynomial form of the adjoint argument, retaining its boundary explicitly. Introduce a function F with $F' = X_1$, and write $p = e^F, q = e^{-F}$. The products below are differential polynomials because the exponential factors cancel. The Lagrange concomitant is

$$L_m = \sum_{j=0}^{m-1} (-1)^j p^{(j)} q^{(m-1-j)}.$$

Termwise differentiation telescopes to

$$DL_m = pq^{(m)} + (-1)^{m-1}p^{(m)}q = (-1)^m A_m(X) - A_m(-X).$$

Equivalently, put $B_j(X) = (-1)^j A_j(-X)$ and write $L_m = \sum_{j=0}^{m-1}(-1)^j B_j(X)B_{m-1-j}(-X)$; then the argument is an identity inside $\mathscr{R}$ and does not require choosing an antiderivative F.

The normal-form projection respects factor degree, and replacing every X_j by $-X_j$ multiplies a degree-ℓ monomial by $(-1)^\ell$. Applying the unique projection to the concomitant identity yields $N_m(-X) = (-1)^m N_m(X)$, proving the parity assertion and (10.10). Finally substitute $A_m = N_m + DT_m$ into (10.9) and use $X_1 DT_m = D(X_1T_m) - X_2T_m$. The result is

$$A_{m+1} = X_1N_m - X_2T_m + D(X_1T_m - N_m - DT_m),$$

which proves (10.11). □

The values of r_m for $m = 2, \dots, 12$ are

$$1, 1, 2, 2, 3, 3, 5, 6, 8, 9, 13.$$

These are formal upper bounds for the number of coordinates in the particular Gamma normal form. No nonvanishing assertion for all allowed coefficients is required. Some low-degree remainders are

$$\begin{aligned} N_4 &= X_1^4 - X_2^2, \\ N_5 &= X_1^5 - 5X_1X_2^2, \\ N_6 &= X_1^6 - 15X_1^2X_2^2 + X_3^2, \\ N_7 &= X_1^7 - 35X_1^3X_2^2 + 7X_1X_3^2, \\ N_8 &= X_1^8 - 70X_1^4X_2^2 + 28X_1^2X_3^2 - 7X_2^4 - X_4^2. \end{aligned} \tag{10.12}$$

In particular the weight-six basis element X_2^3 is excluded by parity. Its absence is a structural consequence, not a fitted numerical cancellation.

10.4 Appell transport of the reduced differential polynomials

Let

$$P_j(Y) = j![t^j]\Gamma(1-t)e^{tY}, \qquad C_m(x) = m![t^m]\frac{\Gamma(1-t)\Gamma(x)}{\Gamma(x-t)}.$$

These are the established scalar and differential Gamma normalizers from the incoming fifth-moment report. The operator ∂_{X_1} commutes with D, and the repeated-largest normal-form space is stable under ∂_{X_1}. Uniqueness gives

$$\partial_{X_1} N_m = mN_{m-1}.$$

Consequently the canonical remainder of C_m is obtained by expanding N_m as a polynomial in X_1 and replacing each power X_1^j by $P_j(\psi)$. Explicitly, with $K_j = N_j|_{X_1=0}$,

$$\mathcal{N}(C_m) = \sum_{j=0}^{m} \binom{m}{j} P_j(\psi)K_{m-j}. \tag{10.13}$$

For example,

$$\mathcal{N}(C_6) = P_6(\psi) - 15P_2(\psi)(\psi')^2 + (\psi'')^2, \tag{10.14}$$

$$\mathcal{N}(C_7) = P_7(\psi) - 35P_3(\psi)(\psi')^2 + 7P_1(\psi)(\psi'')^2. \tag{10.15}$$

The primitive can be obtained by the same finite Gamma coefficient convolution applied to the T_j; constant ambiguities do not affect its endpoint functional.

Moreover, if $C_m = \mathcal{N}(C_m) + D\widetilde{T}_m$, then

$$\mathcal{E}(\widetilde{T}_m) \in \mathbb{Q}[\zeta(2), \dots, \zeta(m-1)]. \tag{10.16}$$

It is homogeneous of weight $m-1$. To see this, put $y = \psi + \gamma$ and use

$$\Gamma(1-t)e^{-\gamma t} = \exp\left(\sum_{j\geq 2} \frac{\zeta(j)}{j} t^j\right).$$

The primitive is a differential polynomial in y with these zeta coefficients. Both $y(1) = 0$ and the constant Laurent coefficient of y at zero are zero, so neither endpoint calculation introduces γ. The finite formula (10.7) proves the assertion at every degree.

10.5 The inherited spectral bridge in the new normal form

Use the conventions

$$\frac{1}{\Gamma(1+u)} = \sum_{j\geq 0} c_j u^j, \qquad \lambda_d = [u]\zeta(1+u, \underbrace{1, \dots, 1}_{d-1}), \quad d \geq 1.$$

The coefficient $[u]$ is a Laurent coefficient, not a derivative at a pole. In particular $\lambda_1 = -\gamma_1$ under the convention $\zeta(1+u) = u^{-1} + \gamma - \gamma_1 u + \cdots$. The established Gamma integral bridge is

$$\int_0^1 C_m(x)\,dx = m!(c_m - \lambda_{m-1}), \qquad m \geq 2. \tag{10.17}$$

Its complete integrand is continuous at zero. Combining this formula with (10.13) and the exact endpoint functional gives an all-degree finite recipe which isolates the single highest height-one coefficient and specifies every remaining mixed integral.

For comparison, the equivalent inverse-Gamma bridge is

$$\mathcal{L}_m := \mathrm{FP}\int_0^1 A_m(x)\,dx = m![t^m]\frac{\sin(\pi t)}{\pi t} - m!\sum_{d=1}^{m-1}(-1)^{m-d-1}c_{m-d-1}\lambda_d. \tag{10.18}$$

This follows directly by multiplying the generating family C_m by $1/\Gamma(1-t)$, retaining $\mathrm{FP}\int C_0 = 1$ and $\mathrm{FP}\int C_1 = \gamma$, and using $1/(\Gamma(1-t)\Gamma(1+t)) = \sin(\pi t)/(\pi t)$. Thus the new normal form changes the finite coordinate list and retains the already established spectral normalization.

11 A direct formula for coincident polygamma products

The manuscript already gives general translated polygamma products in first Hurwitz derivatives, in particular its formula `shj:polyclosurezeta`. The following coincident, original-coordinate finite part has the corresponding ordinary-zeta form. A direct proof is supplied because a collision limit cannot in general be moved through a finite-part prescription without checking its local normalization.

Put $H_0 = 0$ and $H_d = \sum_{j=1}^{d} 1/j$. For nonnegative integers r, s with $d = r + s \geq 1$, define $\sigma_r = (-1)^{r+1}$ and $\sigma_s = (-1)^{s+1}$.

Theorem 11.1 (Coincident bilinear polygamma finite parts)**.** *In the original coordinate x,*

$$\begin{aligned}&\mathrm{FP}\int_0^1 \psi^{(r)}(x)\psi^{(s)}(x)\,dx\\&\quad= d!\left\{(\sigma_r+\sigma_s)\zeta'(d+1)+\left[\sigma_r(H_d-H_s)+\sigma_s(H_d-H_r)\right]\zeta(d+1)\right\}.\end{aligned}\tag{11.1}$$

In particular, for every $r \geq 1$,

$$\mathrm{FP}\int_0^1 (\psi^{(r)}(x))^2\,dx = 2(-1)^{r+1}(2r)!\left\{\zeta'(2r+1)+(H_{2r}-H_r)\zeta(2r+1)\right\}.\tag{11.2}$$

Proof. At the pole $x=-n$, write $v=x+n$ and

$$\psi(-n+v) = -v^{-1}+a_0(n)+\sum_{k\geq 1} a_k(n)v^k,$$

where $a_0(n) = H_n-\gamma$ and $a_k(n) = H_n^{(k+1)}+(-1)^{k+1}\zeta(k+1)$ for $k\geq 1$. In every case $\Delta a_k(n) = n^{-k-1}$. Let $p_j(n)$ denote the coefficient of v^{-j} in the product. There is a constant highest coefficient at order $d+2$, and for $1\leq j\leq d+1$ the other coefficients are

$$p_j(n) = (d+1-j)!a_{d+1-j}(n)\left\{\mathbf{1}_{j\leq r+1}\frac{\sigma_r r!}{(r+1-j)!}+\mathbf{1}_{j\leq s+1}\frac{\sigma_s s!}{(s+1-j)!}\right\}.\tag{11.3}$$

This is just multiplication of each leading pole by the other's analytic Taylor series. In particular

$$\Delta p_1(n) = d!(\sigma_r+\sigma_s)n^{-d-1}.$$

Here are the convergence and normalization details. The normalized Mittag–Leffler expansion is

$$f(z) = P_0(z)+\kappa+\sum_{n\geq 1}\{P_n(z)-P_n(0)\},\qquad P_n(z) = \sum_j \frac{p_j(n)}{(z+n)^j},$$

where κ is the local Laurent constant at zero. Its normal convergence follows from $p_j(n) = O(1+\log n)$ and one subtraction. The whole-circle contour argument eliminates the entire remainder: integrate $z(f(w)-P_0(w))/(w(w-z))$ on $|w| = N+1/2$. Digamma reflection and the differentiated cotangent reflection formulas bound all factors by $O(1+\log N)$ on this circle, so the integral tends to zero. Moreover

$$\kappa-\sum_{n=1}^{N} P_n(0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f((N+1/2)e^{i\theta})\,d\theta \longrightarrow 0.$$

For the last limit, outside an arc of length $O(\log N/N)$ at the negative real axis, at least one differentiated digamma factor tends to zero by the sectorial asymptotic and reflection. On the exceptional arc the uniform logarithmic bound times its length tends to zero. This uses $d\geq 1$; the undifferentiated square is a separate case.

Integrate the normal expansion to one before passing to the cutoff. Finite summation by parts in its higher-pole terms gives

$$\sum_{j\geq 2}\sum_{n\geq 1}\frac{\Delta p_j(n)}{(j-1)n^{j-1}},$$

and the simple poles give $-\sum_{n\geq 1}\Delta p_1(n)\log n$. All terminal terms vanish: $p_j(N)/(N+1)^{j-1}\to 0$ for $j\geq 2$; also $p_1(N)\log(N+1)\to 0$, since it is zero when d is odd, and proportional to the decaying tail $a_d(N)$ when d is even. The grouped sums are absolutely convergent. They are

precisely the original-coordinate finite part, by integration before constant-term extraction at zero.

The logarithmic sum is $d!(\sigma_r + \sigma_s)\zeta'(d+1)$. For the higher poles use the finite identity

$$r! \sum_{h=1}^{r} \frac{(d-h)!}{(r-h)!h} = d!(H_d - H_{d-r}), \qquad 0 \le r \le d. \tag{11.4}$$

After division by $d!$, this identity is obtained by integrating

$$\sum_{h=1}^{r} \binom{r}{h} t^{h-1}(1-t)^{d-h} = \frac{(1-t)^{d-r} - (1-t)^d}{t}$$

from zero to one. The beta integrals on the left and the finite geometric sum on the right give exactly (11.4). Applying it once to each term in (11.3) gives the zeta coefficient in (11.1). Setting $s = r$ proves (11.2). □

The cases needed below are

$$\mathrm{FP}\int_0^1 (\psi')^2\,dx = 2\zeta(3) + 4\zeta'(3), \qquad \mathrm{FP}\int_0^1 (\psi'')^2\,dx = -28\zeta(5) - 48\zeta'(5). \tag{11.5}$$

12 The canonical sixth and seventh digamma identities

Write $z_j = \zeta(j)$ in this section and put $y = \psi + \gamma$. The sixth scalar polynomial is explicitly

$$\begin{aligned} P_6(\psi) = {} & y^6 + 15z_2y^4 + 40z_3y^3 + (45z_2^2 + 90z_4)y^2 \\ & + (120z_2z_3 + 144z_5)y + 15z_2^3 + 90z_2z_4 + 40z_3^2 + 120z_6. \end{aligned} \tag{12.1}$$

Define the one retained mixed moment

$$\widetilde{M}_6 = \mathrm{FP}\int_0^1 (\psi(x) + \gamma)^2(\psi'(x))^2\,dx. \tag{12.2}$$

This is a fully specified original-coordinate finite part, with an ordinary convergent subtracted representation obtained by its finite Laurent principal part. More explicitly,

$$y^2(\psi')^2 = x^{-6} - 2z_3x^{-3} - z_4x^{-2} + 6(z_2z_3 - z_5)x^{-1} + O(1). \tag{12.3}$$

Consequently

$$\begin{aligned} \widetilde{M}_6 = {} & -\frac{1}{5} + z_3 + z_4 \\ & + \int_0^1 \left\{ y^2(\psi')^2 - x^{-6} + 2z_3x^{-3} + z_4x^{-2} - 6(z_2z_3 - z_5)x^{-1} \right\} dx. \end{aligned} \tag{12.4}$$

The complete integrand in braces has a removable singularity at zero.

Theorem 12.1 (A canonical sixth-degree bridge with one mixed moment)**.** *With the conventions already fixed,*

$$\boxed{\begin{aligned} \mathrm{FP}\int_0^1 P_6(\psi(x))\,dx = {} & 720(c_6 - \lambda_5) + 15\widetilde{M}_6 \\ & + 60z_2\zeta'(3) + 48\zeta'(5) + 90z_2z_3 + 270z_5. \end{aligned}} \tag{12.5}$$

Thus this scalar Gamma–Appell combination has only the highest height-one coefficient λ_5 and one specified mixed moment, apart from ordinary constants and ordinary zeta derivatives. This is not an elimination or an independence theorem for $\widetilde{M}_6$.

Proof. For the inverse-Gamma polynomial, the exact primitive from the normal algorithm is

$$T_6 = -3\psi^5 + 20\psi^3\psi' - 15\psi^2\psi'' - 15\psi(\psi')^2 + 6\psi\psi''' + 9\psi'\psi'' - \psi''''. \tag{12.6}$$

Direct differentiation, or the finite recurrence, gives

$$A_6 = \psi^6 - 15\psi^2(\psi')^2 + (\psi'')^2 + T_6'.$$

For C_6, form the polynomial

$$\begin{aligned}\widetilde{T}_6 = T_6[y] &+ 15z_2(-2y^3 + 4y\psi' - \psi'') \\ &+ 40z_3(-\tfrac{3}{2}y^2 + \psi') - (45z_2^2 + 90z_4)y,\end{aligned} \tag{12.7}$$

where $T_6[y]$ means (12.6) with every undifferentiated ψ replaced by y and its derivatives unchanged. Then

$$C_6 = P_6(\psi) - 15(y^2 + z_2)(\psi')^2 + (\psi'')^2 + \widetilde{T}_6'. \tag{12.8}$$

The two endpoint values are

$$\begin{aligned}\widetilde{T}_6(1) &= 52z_2z_3 + 24z_5, \\ \mathrm{CT}_{x=0}\,\widetilde{T}_6(x) &= 112z_2z_3 + 266z_5, \\ \mathcal{E}(\widetilde{T}_6) &= -60z_2z_3 - 242z_5.\end{aligned} \tag{12.9}$$

They follow by substituting $y = -x^{-1} + z_2x - z_3x^2 + z_4x^3 - z_5x^4 + O(x^5)$, or directly from (10.7). Apply finite parts to (12.8), use (10.17), and then use both evaluations in (11.5). Every term in (12.5) follows, including the endpoint contribution and the $z_2\zeta'(3)$ correction. □

For a fully expanded ordinary coefficient,

$$\begin{aligned}720c_6 = \gamma^6 &- 15\gamma^4z_2 + 40\gamma^3z_3 + 45\gamma^2z_2^2 - 90\gamma^2z_4 \\ &- 120\gamma z_2z_3 + 144\gamma z_5 - 15z_2^3 + 90z_2z_4 + 40z_3^2 - 120z_6.\end{aligned} \tag{12.10}$$

12.1 The raw sixth power

Let $Q_6 = \mathrm{FP}\int_0^1 \psi^6$ and $M_6 = \mathrm{FP}\int_0^1 \psi^2(\psi')^2$. The endpoint difference of (12.6) is

$$\mathcal{E}(T_6) = 60\gamma^3z_2 - 150\gamma^2z_3 - 105\gamma z_4 + 210z_2z_3 - 242z_5. \tag{12.11}$$

Combining this with (10.18) gives the complete raw-power formula

$$\begin{aligned}Q_6 = 15M_6 &- 720\lambda_5 + 720\gamma\lambda_4 - 360(\gamma^2 - z_2)\lambda_3 \\ &+ 120(\gamma^3 - 3\gamma z_2 + 2z_3)\lambda_2 \\ &- 30(\gamma^4 - 6\gamma^2z_2 + 8\gamma z_3 + 3z_2^2 - 6z_4)\lambda_1 \\ &- 135z_6 + 48\zeta'(5) - 60\gamma^3z_2 + 150\gamma^2z_3 + 105\gamma z_4 - 210z_2z_3 + 270z_5.\end{aligned} \tag{12.12}$$

The term $-135z_6 = -\pi^6/7$ is the entire contour constant of the sixth inverse-Gamma bridge. It is not an endpoint correction.

12.2 The next odd degree

The algorithm also gives a compact seventh-degree extension. Define

$$W_7 = \mathrm{FP}\int_0^1 P_3(\psi)(\psi')^2\,dx, \qquad K_7 = \mathrm{FP}\int_0^1 P_1(\psi)(\psi'')^2\,dx.$$

Then

$$\boxed{\mathrm{FP}\int_0^1 P_7(\psi)\,dx = 5040(c_7-\lambda_6) + 35W_7 - 7K_7 + 231z_3^2 + 2380z_6.} \tag{12.13}$$

Here $P_1(\psi) = y$ and $P_3(\psi) = y^3 + 3z_2y + 2z_3$. For reproducibility, a primitive for $A_7 - N_7$ is

$$\begin{aligned} T_7 = {} & -\tfrac{7}{2}\psi^6 + 35\psi^4\psi' - 35\psi^3\psi'' - \tfrac{105}{2}\psi^2(\psi')^2 + 21\psi^2\psi''' \\ & + 63\psi\psi'\psi'' - 7\psi\psi'''' + 14(\psi')^3 - 14\psi'\psi''' - \tfrac{21}{2}(\psi'')^2 + \psi''''. \end{aligned} \tag{12.14}$$

In addition to T_6 already displayed, the lower primitives needed for the Gamma convolution are

$$\begin{aligned} T_5 &= -\tfrac{5}{2}\psi^4 + 10\psi^2\psi' - 5\psi\psi'' - \tfrac{5}{2}(\psi')^2 + \psi''', \\ T_4 &= -2\psi^3 + 4\psi\psi' - \psi'', \\ T_3 &= -\tfrac{3}{2}\psi^2 + \psi', \qquad T_2 = -\psi. \end{aligned}$$

The primitive for $C_7 - \mathcal{N}(C_7)$ is

$$\begin{aligned} \widetilde{T}_7 = {} & T_7[y] + 21z_2T_5[y] + 70z_3T_4[y] \\ & + (105z_2^2 + 210z_4)T_3[y] + (420z_2z_3 + 504z_5)T_2[y]. \end{aligned}$$

The finite endpoint algorithm gives

$$\mathcal{E}(\widetilde{T}_7) = -448z_2z_4 - 231z_3^2 - 1596z_6 = -231z_3^2 - 2380z_6.$$

Combining it with (10.15) and the inherited Gamma bridge proves (12.13). The certificate contains the full polynomial identity, so this seventh-degree formula does not rely on a numerical extrapolation from degree six.

12.3 Proof status, verification, and remaining question

The accompanying exact program checks every partition at every weight through twelve: it verifies the decomposition identity, the vanishing normal form of every total derivative, the bidegree count, the adjoint concomitant identity, and the parity restriction. It also independently expands the local Laurent polynomials for the displayed sixth and seventh endpoint constants. These finite checks corroborate the proofs; the terminating algorithm and the dimension theorem themselves hold at all degrees by the arguments above.

This resolves the formal integration-by-parts classification requested in the incoming programme and gives complete degree-six and degree-seven applications. The next separate task is to determine the image of this formal quotient in an explicitly specified family of harmonic spectral coordinates. Its kernel may contain additional special-function relations. Neither the partition count nor the parity count can be used to assert arithmetic independence of the remaining constants.

13 Source audit, proposed status changes, and further research

13.1 What should change in the existing programme

The audit was directed at the statements used or continued in this article. It was not a line-by-line certification of every chapter in the large collective manuscript. No mathematical counterexample or invalid proof step was established in those source statements. The principal proposed changes are strengthened theorems and more precise boundaries between formal and arithmetic conclusions.

1. **Split the optimal-prefix conjecture by scope.** In [2, `conj:firstq`], replace the $p = 2$, $p = 4$, and $p = 6$ parts by Theorems 3.1, 5.1, and 6.2, including their determinants and Corollary 3.4. Retain the diagonal theorem already proved there. Theorem 4.3 gives an explicit fixed-size reduction for all even pairs. Leave unequal even pairs $8 \le p < q$ as the remaining conjecture. Finite checks at those orders should continue to be identified as checks, with no inferred all-order conclusion.

2. **Extend the directional-zeta chapter through cubic order.** Append Theorem 8.1 and the two explicit cancellations (8.16) and (8.17) after the inherited quadratic theory. Preserve the precise admissible-ray condition and the displayed polar subtraction. A global third Taylor tensor of T at the origin is not defined; the theorem concerns specified one-variable restrictions. The kernel theorem should be attached to the exact local restrictions it classifies.

3. **Replace the unstructured higher-power task by a proved algorithm and a smaller arithmetic task.** Theorems 10.1 and 10.2 settle the formal integration-by-parts classification in the chosen differential-polynomial space. Equations (12.5) and (12.13) give explicit applications. What remains is the spectral evaluation of their specified mixed moments, not the existence of an integration-by-parts procedure.

4. **Preserve the two finite-part normalizations.** Centered tail moments use continuation of $D_{p,q}(s)$, while singular integrals use the constant term in the original coordinate x. The proofs state why the particular ordinary subtractions recover those values. An unannounced translation of the cutoff or omission of $\mathcal{E}(T)$ could alter constants. The special binomial omission rule in (2.7) must also remain explicit: restoring a forbidden lower-index-zero term changes the matrix under discussion.

5. **Keep the priority and independence statements narrow.** The Bernoulli Hankel determinant and differential-polynomial exactness machinery have classical antecedents. The height-one coefficient bridge is inherited. The new content lies in the mixed determinant, the resulting exact cancellations, the cubic directional reduction, and the explicitly normalized higher-power applications. Formal coordinate counts do not prove independence of zeta values or of the remaining mixed integrals. No historical priority assertion beyond the inspected material is needed for these mathematical contributions.

The manuscript's short S_6 identity and its revised S_8 candidate remain conjectural. The older rejected S_8 vector is a different statement; it must not be substituted for the current candidate. The present results neither prove nor refute those period identities [1, README and editorial ledger].

13.2 A proposed programme of exact research questions

The following questions are selected for identities, reduction theorems, and explicit primitives. They are ordered approximately by their connection to the results above, rather than by an unproved prediction of difficulty.

R1. Prove the remaining unequal-order prefix conjecture. For $p = 2a < 2b = q$ with $a \geq 4$, use the explicit a-square matrix in Theorem 4.3 to finish the remaining nonvanishing problem. The shifted-Hankel part is already known to be nonsingular, and the weighted Bernoulli reduction has degree at most $2a - 1$. The next task is to evaluate the residual determinant and find an arithmetic obstruction to its vanishing, as in the $a = 1, 2, 3$ proofs. The exact $(a, b) = (4, 9)$ example in Section 6 rules out the naive uniform valuation $-a$; any more general argument must accommodate that obstruction. A proof must control nonvanishing of the final rational factor, not just display a product of unknown minors.

R2. Determine every reconstruction row in closed form. Theorem 3.3 gives the rational coordinate Z_0. An explicit formula for every row of M_b^{-1} would express each even zeta coordinate through the same optimal moment prefix. Orthogonal polynomial kernels followed by the rank-one inverse formula offer a concrete route. Seek coefficient formulas with finite binomial sums and prove their normalization directly. The result would produce systematic ordinary harmonic-tail identities with prescribed $\zeta(2k)$ values.

R3. Extend the rational tail cancellations to shifted or character-weighted tails. Replace the ordinary tails by Hurwitz tails with a rational shift, or by tails weighted by Dirichlet characters. Determine which centering and parity permit an exact polynomial subtraction, and which generalized Bernoulli determinants replace M_b. The half-integer vanishing $\zeta(-2r, 1/2) = 0$ is a special ingredient of the present proof; in a new shift it must be replaced by an explicitly computed finite correction.

R4. Find finite polylogarithmic evaluations associated with the boundary family. Equation (7.3) is an exact endpoint identity for an explicit finite differential operator. Determine whether its numerator admits useful integrated forms at interior algebraic arguments, with depth reduction proved by differential identities and base values. The test is an equality of functions or a rigorously anchored primitive, not a numerical integer relation at a single argument.

R5. Determine the arithmetic meaning of the second cubic Tornheim coordinate. The constant κ is fixed by $J_{AA}(0)$ in the canonical subtraction, and (8.13) already expresses it through the strict Euler double-zeta ray $\zeta(2t, t)$. Starting from that proved representation, compare it with the inherited cubic digamma invariant and seek a reduction. The concrete question is whether κ belongs to a specified algebra generated by ordinary zeta derivatives and Ω, or whether a further higher-depth coordinate is required. A coordinate change alone does not answer that question.

R6. Compute the quartic and quintic ray cancellation laws. Theorem 8.4 gives formal ambiguity dimensions 3 and 6 at degrees four and five. Construct explicit homogeneous direction functions for these spaces, derive the ordinary-zeta inhomogeneous part, and find short integer-weighted combinations of positive rays that annihilate all undetermined coefficients. These are finite symbolic problems once the analytic polar normalization has been fixed, but the coefficient calculation must be independently checked.

R7. Add further Tornheim functional relations to the local kernel and compute the exact reduction. The kernel theorem deliberately uses a specified subset of the known relations. Linearize additional functional equations from the classical Tornheim literature at the same origin, keeping their Gamma and trigonometric factors to the required order. Determine whether they reduce the cubic two-coordinate space or only change its basis. At higher degrees, compute the quotient of the kernel by these additional restrictions before making claims about necessary arithmetic data.

R8. Evaluate the retained sixth mixed moment in an explicitly stated spectral family. The quantity $\widetilde{M}_6$ has a convergent subtracted integral in (12.4). Use its pole principal

parts, rather than numerical fitting, to express it through ordered harmonic zeta derivatives. Then test whether the resulting combination reduces within a fixed family of height-one, depth-two, or Tornheim coordinates. The desired theorem is an explicit evaluation or a proved reduction in that family; the present formula makes no assertion that $\widetilde{M}_6$ is irreducible.

R9. Determine the exact image of the differential quotient. The partition theorem classifies formal integration by parts. Define the map from this quotient, together with the endpoint functional, to the harmonic spectral coordinates supplied by the pole expansion. Compute its kernel at each weight and identify structural generators for the additional relations. The square formula is an example of a relation available after substitution into the digamma function that is not an identity in the free differential-polynomial quotient.

R10. Combine higher Stieltjes coefficients with the normal form and parameter primitives. The fixed Mellin family in Proposition 9.1 gives every $\lambda_{d,j}$ and ordinary γ_j at depth one. Introduce the appropriate logarithmic parameter derivatives into the Gamma family, determine their differential normal forms, and retain the complete endpoint terms. A useful target is a closed recurrence for primitive families whose spectral derivatives isolate one prescribed $\lambda_{d,j}$, followed by analogous formulas for Hurwitz $\gamma_j(a)$ with a fixed shift convention.

R11. Explain the coefficient patterns of the Gamma normal forms. The Appell property already proves, for example, that the coefficient of $X_1^{m-4}X_2^2$ is $-\binom{m}{4}$: its undifferentiated core is the $-X_2^2$ term in N_4. The remaining coefficient problem is therefore concentrated in $K_j = N_j|_{X_1=0}$. Prove general formulas for families of repeated-largest partitions in these cores using the finite recurrence. Distinguish allowed parity coordinates from coordinates whose coefficient actually vanishes; an all-degree support theorem would sharpen the current formal upper count.

R12. Return to the short Gaussian identities with a fixed period target. For S_6 and revised S_8, retain the repository's exact frozen vectors and the existing shuffle equivalences. Seek a new functional or iterated-integral identity whose boundary evaluation is exactly that vector. A nonzero projection in a chosen formal relation space does not refute the numerical period identity, and a small computed residual does not prove it. The new determinant and differential methods may suggest useful auxiliary identities, but no direct implication is established here.

Each question specifies the missing mathematical step. In particular, the status of a constant evaluation should change only when an analytic identity or an exact algebraic reduction with justified boundary values has been supplied.

14 Reproducibility and the limits of computation

The accompanying package contains the complete typeset article, a single flattened TEX source, the modular sources, eleven verification programs, their recorded JSON results, and an integration guide. The proofs in the article establish the uniform statements. The programs provide exact finite certificates and independent numerical diagnostics at explicitly specified orders.

14.1 Exact finite certificates

The order-two verifier constructs the matrix directly from the inherited binomial formula, without defining it by the proposed factorization. For every $b = 2, \ldots, 14$ it checks the determinant, the

rank-one factorization, the Padé inverse-row contraction, the rational extraction row, the shifted Hankel product, and the final factorial cancellation. It also checks the Padé coefficient identities through the corresponding orders and reconstructs the ordinary-series examples at $b = 2, 3, 4$.

The order-four verifier checks the symbolic perturbation contractions, the resulting determinant polynomial, and its exact parity calculation. It also constructs the original source matrices for $b = 3, \dots, 14$, including the exceptional smallest pair $(4, 6)$. The general shifted-Hankel verifier checks all 45 pairs $1 \le a \le b \le 9$, comparing the source entries with the factorization, the determinant signs, the size-a reduction, and the weighted Bernoulli sums. The exact ranges supplement the uniform proofs of both new sections.

The order-six program constructs the normalized 3×3 matrix with an indeterminate b. It proves the determinant normalization, constructs its numerator in the integer polynomial ring, and verifies the modulo-four specialization used in the nonvanishing proof. Direct matrices are also checked at $b = 4, \dots, 12$, and the order-eight counterexample to the uniform valuation pattern is computed exactly. A separate short program checks the explicit $(4, 6)$ ordinary tail identity. These computations are finite polynomial certificates, not interpolation from integer samples.

The Tornheim symbolic verifier multiplies the Gamma and zeta Taylor series in exact rational arithmetic. It verifies both elementary restrictions on J, the full cubic direction formula, and both printed cancellation identities. A separate construction builds the linear restriction matrices for homogeneous polynomial germs through degree ten and checks the stated kernel dimensions. An additional independent script derives the three-ray cancellation directly from raw series and the two elementary constraints; it also detects a deliberately changed coefficient.

The differential verifier checks every partition monomial at every weight through twelve. It verifies its exact decomposition into normal form and derivative, the zero normal form of every derivative column, the factor-degree counts, the adjoint concomitant, and the Gamma parity. It constructs the full degree-six and degree-seven polynomial identities, including the canonical endpoint constants. The Laurent principal part of the retained sixth mixed integral is checked symbolically as well.

The checks over finite ranges confirm the formulas within those ranges. They do not by themselves prove the uniform statements. The identities certified with indeterminate b, including the integer-polynomial certificate in Section 6, are exact finite algebraic steps in the proofs. The uniform conclusions also use the Padé coefficient argument, the Bernoulli congruences, holomorphic divisibility, and the terminating partition argument, as applicable. No external proof assistant was used to formalize the article's analytic proofs.

14.2 Independent numerical diagnostics

Table 2 reports the main recorded comparisons. Every entry is floating-point evidence, not an interval enclosure. The numerical evaluators do not assume the identities they are checking.

For the harmonic sums, the exact polynomial subtraction removes every nonnegative asymptotic power; the remaining inverse powers are summed by ordinary Hurwitz zeta values to accelerate a convergent series. For the independent $(2, 8)$ check, increasing the precision and the tail parameters changed the residual from approximately $4.20 \cdot 10^{-52}$ to $3.28 \cdot 10^{-72}$.

The Tornheim evaluator integrates the Jonquière expansion below $x = 1$ and sums an incomplete-Gamma tail above $x = 1$. With 50-digit arithmetic, series index 48, tail index 110, and seven odd-part steps starting at $h = 0.0008$, it gives

$$W'''_{1,2,3}(0) + 4W'''_{2,3,1}(0) - 5W'''_{3,1,2}(0) \approx -451.35351956848625606446741456754 05.$$

The code does not use the proposed directional formula, fit Ω or κ, or derive its values from the digamma moment.

Table 2: Recorded numerical comparisons; residuals are diagnostic.

Identity	Independent evaluation	Absolute residual
Ordinary $(2,4)$ tail sum	100-digit tails; 100 initial terms and 24 asymptotic terms	$9.93 \cdot 10^{-70}$
Ordinary $(2,6)$ tail sum	Same method, independently formed weight polynomial	$3.20 \cdot 10^{-65}$
Ordinary $(2,8)$ tail sum	100 digits; 120 initial terms and 32 asymptotic terms	$3.28 \cdot 10^{-72}$
Three cubic Tornheim rays	Subtracted Mellin evaluation and odd-part extrapolation	$6.30 \cdot 10^{-33}$
Canonical sixth digamma formula	Direct Laurent subtraction and separate height-one Mellin integral	$5.63 \cdot 10^{-51}$
$(\psi')^2$ and $(\psi'')^2$	Direct Laurent subtraction and ordinary quadrature	$1.23 \cdot 10^{-90}$, $1.77 \cdot 10^{-88}$

For the sixth identity, direct finite-part quadrature gives

$$\mathrm{FP}\int_0^1 P_6(\psi(x))\,dx \approx 20.822613154257822426734055105133 3,$$
$$\widetilde{M}_6 \approx -3.402445086569186617474980174480 45.$$

The separate Mellin integral uses an explicit lower cutoff 10^{-60}; the omitted piece is the limiting numerical effect in the displayed comparison. Its size was not promoted to a certified error bound. The formula itself follows from exact polynomial identities, exact endpoints, and the proved ordinary Gamma bridge.

14.3 Build and integration procedure

Run `pythonbuild.py` to regenerate the flattened source and PDF. The build uses `pdflatex` and resolves cross-references in repeated passes. Run `pythonverification/verify_all.py` for the exact certificates. The optional flag `--numerical` also runs the numerical diagnostics; the Mellin computation is the longest job. Required Python packages and the versions used are recorded in the package. The JSON files contain parameters and full recorded output, so a future replay can distinguish a change in arithmetic, a change in numerical settings, and a change in the mathematical claim.

The integration guide maps each contribution to its source question and names the labels that may be imported into the manuscript. It proposes status edits without changing the pinned source checkout. The article is intended to be read and checked as a research continuation before those edits are adopted.

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[2] ProveIt incoming research, *Canonical Digamma and Fifth Moment*, 11 October 2026. Archive `docs/incoming/ProveIt_Canonical_Digamma_Fifth_Moment_2026-10-11.zip` at the pinned repository commit [1]; especially `sections/02_laurent.tex`, `03_gamma.tex`, `07_tail_rank.tex`, and `09_questions.tex`. The archive is the mathematical baseline for the optimal-prefix conjecture and the Gamma integral normalization used here.

[3] ProveIt incoming research, *Higher Digamma Powers and Ordered Gamma Sums*, 11 October 2026. Archive `docs/incoming/ProveIt_Higher_Digamma_Powers_2026-10-11.zip` at the pinned repository commit [1]; especially its sections on differential polynomials and further exact research questions.

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